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Business Buzz

Final Documentation By BuzzTech Solutions

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Table of Contents

1. Business case *1*

Entrepreneurs Struggle to Navigate Fragmented Business Registration and Compliance Processes **1**

1.1 The problem 1
1.2 Project objectives 3
1.3 Problem background 4
1.4 Related systems analysis 7
1.5 The project plan 13
1.6 Risk Analysis 19

2. System Requirements *22*

Specifications and Technical Design **22**

2.1 Plan to Address Objectives 22
2.2 Project scope 24
2.3 Business requirements 28
2.4 Hardware and Software Requirements 31
2.5 Design constraints 32
2.6 High-level use case diagram 35
2.7 UML Class Model / Diagram 39
2.8 Relational Database Model Diagram 40
2.9 User Interface Design 41
1.9.2 Approver Dashboard 46

3. Implementation **49**

3.1 Choice of Tools 49
3.2 Extracts of complex code 52
3.2.2 Business Profile Creation and Validation 53
3.3 Source code references (if applicable) 55

3.4 Problems encountered **58**

3.4.1 Creating an Web Application that has an AI Asistant 58
3.4.2 Persistence 58
3.4.3 Developing a methodology to my coding 59

4. Conclusion *60*

References *61*

A. Business case revisions *62*

A.1 Changes made in response to comments **62**

A.1.1 Justify (align) document text 62

A.2 Changes made as the project progressed **62**

A.2.1 Changed objectives 62

B. System requirements, specifications and technical design revisions *64*

B.1 Changes made in response to comments **64**

B.2 Changes made as the project progressed **64**

B.2.1 Extra feature added 64

C. Implementation revisions *65*

C.1 Changes made in response to comments **65**

C.1.1 Extract of complex code was boilerplate 65

1. Business case

Entrepreneurs Struggle to Navigate Fragmented Business Registration and Compliance Processes

1.1 The problem

Entrepreneurs Struggle to Navigate Fragmented Business Registration and Compliance Processes

What the Problem Is
Starting a business in South Africa is not simply a matter of having a good idea. Before an entrepreneur can legally operate, open a business bank account, apply for funding, or trade with larger organisations, they must complete a complex series of registration and compliance processes spread across multiple government institutions.
The real-world problem is that first-time and small-business entrepreneurs cannot easily determine what they need to do, in what order, and with which institution. Information is scattered across different government websites, each explaining only its own services. Entrepreneurs must piece together the full picture themselves - often discovering missing documents or incorrect information only after applying. This results in delays, additional costs, and in many cases, abandonment of the formalisation process altogether.

This is not a problem of missing software. South Africa already provides digital services for individual parts of the journey. CIPC eServices provide company registration and transaction-status functionality, BizPortal consolidates several business-related services, and SARS eFiling supports tax returns and payments (Companies and Intellectual Property Commission [CIPC], South African Revenue Service [SARS], 2025). The problem is that these services are fragmented, uncoordinated, and do not guide the entrepreneur through the complete journey.

The Domain
The problem occurs within the South African business registration and compliance environment. This domain involves multiple government departments and agencies, each with its own systems, requirements, and processes:

Institution	Responsibility
Companies and Intellectual Property Commission (CIPC)	Company registration, annual returns, director amendments
South African Revenue Service (SARS)	Tax registration, returns, declarations, payments
Department of Employment and Labour (UIF)	Unemployment Insurance Fund registration for employers
Department of Home Affairs (DHA)	Identity verification for company directors and members

Municipal authorities	Business licences and permits
Institution	Responsibility

While individual digital services exist, there is no single platform that coordinates the entrepreneur's journey across all of them. The entrepreneur is left to navigate this fragmented landscape alone.

Who Encounters the Problem

The problem is encountered by first-time and small-business entrepreneurs across South Africa, particularly:

- First-time entrepreneurs with little or no experience of formal business registration
- Informal business owners seeking to formalise their operations
- Small-business owners in rural and underserved communities where access to business advisory services is limited
- Entrepreneurs without access to expensive private consultants who could otherwise guide them through the process

These individuals often rely on internet searches, social media, friends, or expensive consultants to understand what is required. For many, the complexity and cost of navigating the system is a significant barrier to formalisation — which in turn limits their access to funding, banking services, government tenders, and legal protection.

Why This Matters

SMMEs employ approximately 59% of South Africa's workforce (Organization for Economic Co-operation and Development [OECD], 2025), making them essential to employment and economic participation. Yet regulatory and administrative burdens place disproportionate pressure on smaller firms, which have fewer resources available for compliance activities than larger organisations.

Research identifies regulatory and legislative requirements as significant inhibitors of SME establishment and growth. Research on tax compliance further identifies knowledge gaps, communication difficulties, resource limitations, and compliance costs as key challenges.

When entrepreneurs cannot navigate the registration and compliance journey, the consequences are:

- Delays in becoming legally operational
- Additional costs from rejected applications and resubmissions
- Lost economic opportunities - including funding, tenders, and partnerships
- Abandonment of formalisation - with businesses remaining informal and outside the tax and regulatory net

Documentation Burden

Documentation adds a further layer of administrative effort. Entrepreneurs may need to provide identity information, business information, proof of address, and other supporting information - often repeatedly across different applications. This increases the risk of omissions and inconsistencies and adds to the time and cost burden of compliance.

Conditional Compliance Requirements

Some compliance requirements depend on the entrepreneur's specific circumstances:

- UIF registration is relevant only to employers who employ people for remuneration — it is not a universal requirement (Unemployment Insurance Fund [UIF])
- Tax and registration responsibilities differ according to the type and structure of the business

This means a one-size-fits-all approach to guidance is insufficient. Entrepreneurs need guidance that adapts to their specific situation.

1.2 Project objectives

The main objective of Business Buzz is to design and develop a web-based centralised business registration and compliance management system that simulates selected institutional processes and coordinates the entrepreneur's journey from business registration through relevant compliance stages and a simulated banking assessment.

1. Centralise the business registration and compliance journey by presenting relevant stages through one coordinated platform.
How it will be done: A single web-based platform will consolidate CIPC, SARS, UIF, and Home Affairs-related stages into one coordinated workflow, accessible through a single authenticated account.

2. Implement three primary role-based environments: Entrepreneur, Approver and Bank.
How it will be done: Role-based access control (RBAC) will provide distinct environments for each role. Entrepreneurs will manage their business journey, Approvers will review applications, and Bank users will review simulated compliance information.

3. Create simulated institutional entities representing selected CIPC, SARS, UIF, and Home Affairs processes.
How it will be done: Internal simulation modules will model selected processes for each institution using controlled internal logic rather than live government APIs. All outputs will be clearly labelled as simulated.

4. **Capture entrepreneur and business information once and reuse it across applicable workflow stages.**
How it will be done: Information will be stored in a central relational database and retrieved when needed at later stages, eliminating repeated data entry.

5. **Generate digital documents automatically from information captured within the system.**
How it will be done: A document generation service will populate predefined templates using stored profile information and generate PDF documents on demand.

6. **Provide print-ready hardcopy versions of generated documents where a physical document is required within the simulated workflow.**
How it will be done: All documents will be formatted for A4 printing with professional layouts, include a "SIMULATED" watermark, and be available for download or direct printing.

7**. Implement an approval workflow that allows the Approver to review, approve, reject, or request further information at defined stages.**
How it will be done: The Approver role will have access to a review dashboard. Approvers can inspect submitted information, record a decision, and provide comments that update the application status and notify the entrepreneur.

8. **Provide simulated real-time status updates when workflow actions occur within the Business Buzz environment.**
How it will be done: Application and stage statuses will update immediately when workflow actions occur. In-system notifications will alert users of significant events.

9. **Provide a centralised compliance view showing completed, pending, rejected and outstanding stages.**
How it will be done: The Entrepreneur Dashboard will display all applicable stages with status indicators and show overall progress percentage and a summary of completed versus outstanding stages.

10**. Allow the Bank role to review relevant simulated registration and compliance information during a simulated loan or financial-service assessment.**
How it will be done: The Bank role will access a dashboard showing businesses submitted for assessment, review permitted information, and record a simulated assessment outcome.

11**. Implement role-based access control so users can access only the information and functions relevant to their roles.**
How it will be done: Server-side role-based authorisation will be enforced using ASP.NET Core Identity. Unauthorised access attempts will be redirected.

12. **Maintain an auditable record of important workflow actions, including submissions, approvals, rejections, and status changes.**
How it will be done: An audit trail will record the user responsible, the action performed, the affected record, and the date and time for each significant action.

13. **Evaluate the system through functional, usability, and security testing before final presentation.**
How it will be done: The system will undergo unit testing, integration testing, usability testing, and security testing. Results will be documented as evidence of system quality.

1.3 Problem background

1.3.1 Focal Point: Micro and Very Small Enterprises in the Pre-Registration Phase

Business Buzz focuses specifically on **micro and very small enterprises** that are in the **pre-registration or early-registration phase** of their business journey.

According to the National Small Business Act 102 of 1996, micro enterprises employ between 0 and 5 people, while very small enterprises employ between 6 and 20 people (Republic of South Africa, 1996). These businesses typically:

- Operate informally or are in the process of formalising
- Have limited resources for administrative and compliance activities
- Have little or no experience with formal registration processes
- May or may not have employees (affecting UIF applicability)
- Rely on the owner to manage all aspects of the business

This focal point limits the scope of Business Buzz to a specific, well-defined segment of the SMME sector. It ensures that the system's conditional workflow logic, document generation, and simulated processes are relevant to the target users. It also makes the project feasible within the academic timeframe by excluding larger, more complex businesses with established compliance structures.

1.3.2 The Economic Importance of SMMEs in South Africa

Small, Medium and Micro Enterprises (SMMEs) are widely recognised as a critical driver of economic growth, employment, and poverty reduction in developing economies. In South Africa, the OECD reports that micro, small and medium-sized enterprises employed approximately **59% of the country's workforce in 2022** (Organization for Economic Co-operation and Development [OECD).

Within this sector, micro and very small enterprises represent the largest proportion of businesses and are the most likely to face barriers to formalisation. Their contribution makes the conditions under which smaller businesses enter, operate, and grow relevant to employment and economic development.

1.3.3 Regulatory and Legislative Challenges Facing SMMEs

The regulatory environment is a recurring theme in research on South African SMEs. identified the regulatory environment, legislation, and compliance as important inhibitors of SME establishment and growth. The study highlighted specific challenges associated with:

- Labour laws
- Taxation requirements
- Legal compliance obligations
- Municipal regulations

Similarly identified regulatory and legislative compliance, alongside limited resources, as significant challenges affecting SMEs. These findings suggest that the regulatory burden is not merely a minor inconvenience but a substantial barrier to business formation and growth — particularly for micro and very small enterprises with limited capacity.

1.3.4 Regulatory Complexity and the Need for Clear Information

The complexity of regulatory requirements is a particular concern for smaller firms. The OECD's 2025 Economic Survey identified **red tape** as a specific challenge for South African SMEs and noted that communicating and simplifying regulatory requirements can help entrepreneurs understand their obligations and reduce compliance costs.

The OECD's 2026 assessment further identifies regulatory complexity as a constraint for SMEs, which have less capacity to deal with complex requirements than larger firms. The report recommends clearer language and better organisation of business permits and licences.

These findings are directly relevant to Business Buzz. The proposed system is designed to organise a complex multi-stage journey into a structured workflow progression for micro and very small enterprises. It does not attempt to replace official regulatory sources; rather, it models selected processes and uses clearly labelled simulation outputs to demonstrate how information could move between stages.

1.3.5 Tax Compliance and Information Challenges

Tax compliance provides a further example of the information and preparation challenges that affect SMEs. Myburg and Schutte (2025) studied SME owners and managing directors in the Northwest Province and identified several themes affecting tax compliance:

- Knowledge gaps
- Communication deficits
- Resource limitations
- Compliance costs

While the study is geographically limited and should not be generalised to every South African entrepreneur, it provides relevant evidence that understanding tax obligations and accessing appropriate information can be difficult for some small-business owners. This is particularly true for micro and very small enterprises that lack dedicated administrative staff.

1.3.6 Existing Digital Government Services

South Africa has established digital services for important business and compliance processes. *CIPC eServices* provide electronic access to company-related transactions, while BizPortal provides a centralized digital channel for company registration and related services. SARS eFiling provides electronic interaction for tax returns, declarations, payments, and other tax-related activities. CIPC also documents identity verification involving the Department of Home Affairs in relevant processes. CIPC states that directors or members can be verified against Home Affairs information in real time for selected company-registration processes. This provides a factual basis for modelling a Home Affairs identity-verification stage within *Business Buzz*, while the Business Buzz implementation itself remains a simulation rather than a live DHA integration.

1.3.7 UIF and Conditional Compliance

UIF-related obligations illustrate why the proposed system should use conditional workflows. The Department of Employment and Labour states that employers who employ people for remuneration must register with the UIF and make the required declarations and contributions. The official uFiling service supports electronic UIF declarations and contributions (Unemployment Insurance Fund .

This means that UIF registration is not a universal requirement for every business. It applies only to employers. For micro and very small enterprises, some may have employees while others may not. A system that presents every entrepreneur with the same list of requirements would therefore be inaccurate and potentially misleading. Business Buzz will therefore simulate the UIF stage only when the entrepreneur's business circumstances indicate that the stage is applicable.

1.3.8 Multi-Stakeholder Coordination as the Project Opportunity

The existing literature and official systems indicate that the business registration and compliance environment contains a maze of multiple responsibilities, institutional processes, and information requirements.
The project opportunity identified by Business Buzz is therefore **not to replace existing government systems**, but to demonstrate a centralised workflow that coordinates selected institutional processes from the perspective of a micro or very small enterprise entrepreneur.
The proposed system extends the preparation concept into a multi-stakeholder simulation. An entrepreneur can:

1. Create a business profile
2. Progress through simulated Home Affairs and CIPC stages
3. Generate documents
4. Move through approval points
5. Proceed to simulated SARS and UIF stages where applicable
6. Expose relevant compliance information to the Bank role for a simulated financial assessment

This represents the project's own proposed system design based on the problem and related-system analysis.

1.3.9 Literature Review Summary

The reviewed literature indicates that:

- SMEs, particularly micro and very small enterprises, are important to South Africa's economic and employment environment
- Regulatory complexity, administrative burdens, and tax-compliance challenges create difficulties for smaller firms
- Existing digital platforms demonstrate that many individual services are already available online
- However, these services are fragmented and do not provide coordinated guidance across the complete journey
- Micro and very small enterprises in the pre-registration phase are the most affected and the least equipped to navigate this complexity

The identified opportunity for Business Buzz is to demonstrate how selected registration, compliance, documentation, and approval processes could be coordinated through one centralised information system for this specific target segment, while clearly distinguishing simulated functionality from official institutional transactions.

1.4 Related systems analysis

Three existing web-based systems were analyzed because they provide services or workflows that overlap with the business-registration and compliance journey addressed by Business Buzz. The analysis identifies useful functionality to adapt, functionality that should not be duplicated, and the distinctive position of Business Buzz. The supplied Business Case template requires at least three related systems, platform descriptions, screenshots, figure captions, features to adapt, features to avoid, and references.

1.4.1 CIPC eServices

Platform: Web

CIPC eServices is an online CIPC platform that provides company-related services, including company registration, director amendments, annual returns, name reservations, and transaction-status functionality.

Figure 1.1 demonstrates the CIPC eServices interface and illustrates how customer access, registration, and company-related services are organised within the platform.

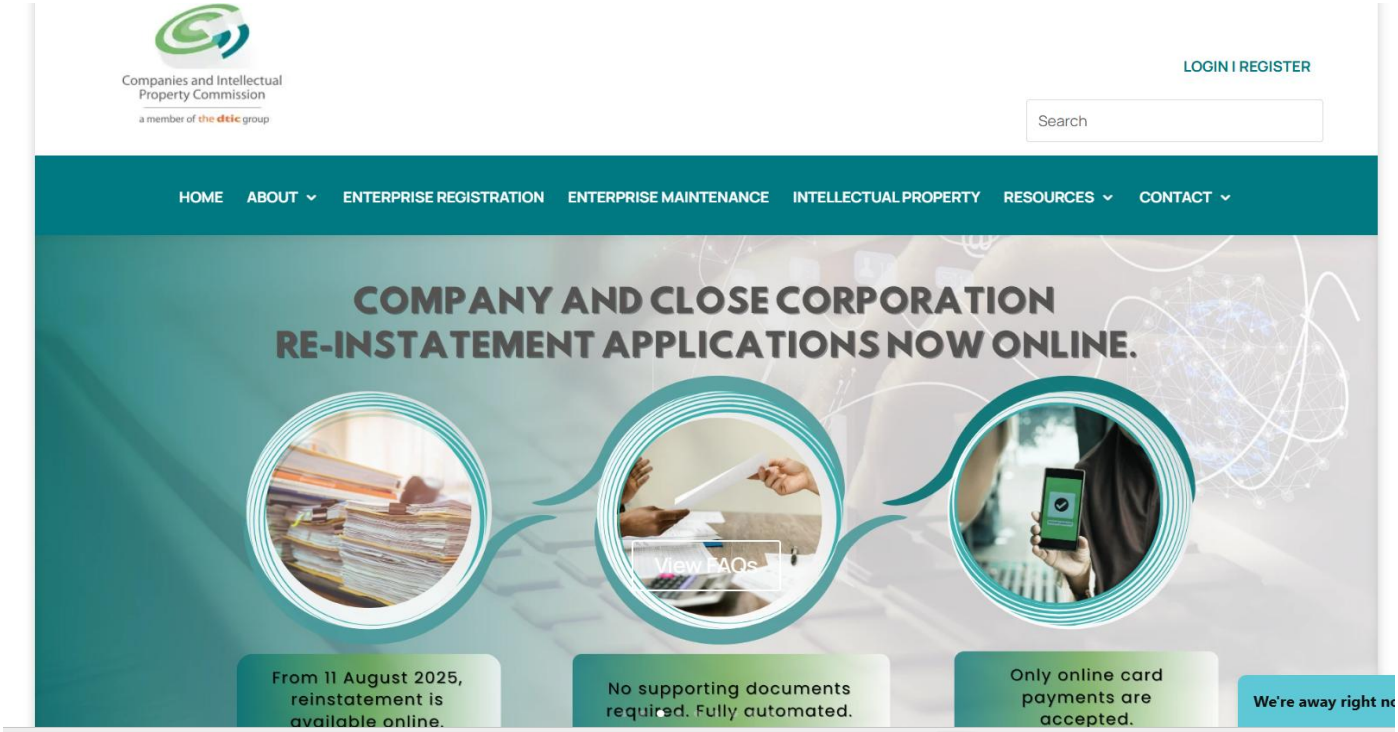


Figure 1.1: Source: CIPC eServices website.

Features/processes to adapt:

- Secure customer registration and login.
- Structured service navigation.
- Transaction or task status tracking.
- Electronic handling of company-related information and documents.
- Authentication and access controls.

Features/processes to avoid or improve:

- Do not reproduce the full CIPC transactional system.
- Do not present Business Buzz outputs as official CIPC approvals or certificates.
- Do not depend on live CIPC integration for the core academic prototype.

Relevance to Business Buzz

CIPC eServices are relevant because it demonstrates how an official government system allows entrepreneurs to perform company-related transactions online. However, Business Buzz would not replace CIPC eServices. Instead, Business Buzz would provide preparation and guidance before the user proceeds with the relevant official service.

1.4.2 CIPC BizPortal

Platform: Web

BizPortal is a CIPC platform developed to provide company registration and related services through a centralized digital environment. Its service description includes company registration, tax registration, UIF registration, Compensation Fund registration, and business bank account services, among others.

Figure 1.2 demonstrates how BizPortal consolidates several business-related services within one platform. This centralization is relevant to Business Buzz because it supports the decision to use one central hub rather than separate user journeys for every simulated institution.

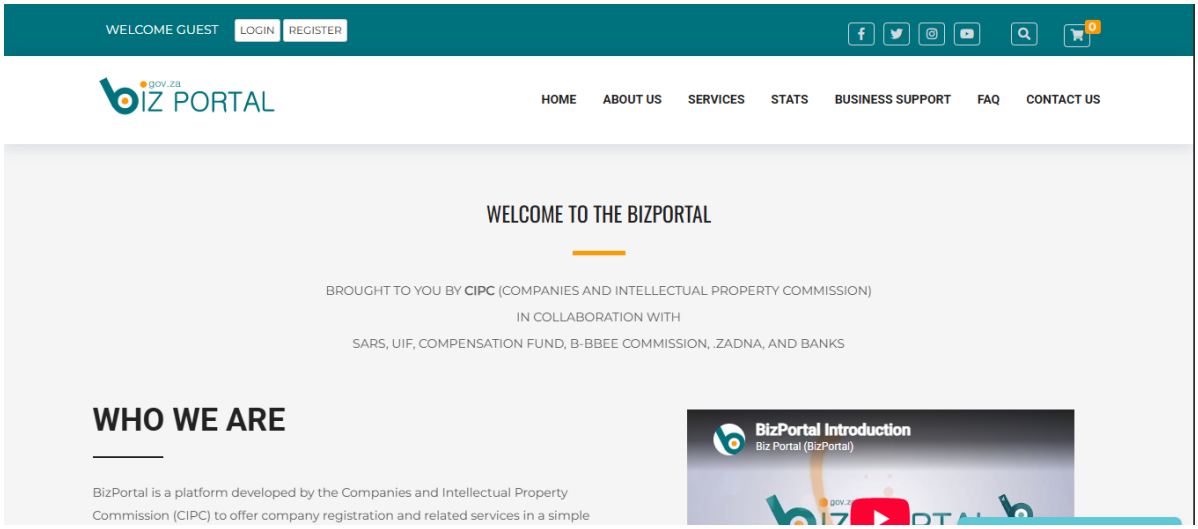


Figure 1.2: CIPC BizPortal interface

Features/processes to adapt:

- Centralized access to related business services.
- Grouping of multiple business-support processes.
- Simple digital workflows.
- Business information reused across related services.
- A single-entry point for a multi-stage business journey.

Features/processes to avoid or improve:

- Business Buzz should not claim to be another official government portal.
- All simulated institutional decisions and statuses must be clearly labelled.
- Implement only the selected processes needed to demonstrate the agreed scope.

Relevance to Business Buzz

BizPortal is particularly relevant to Business Buzz because it already brings several business-related services together. However, the proposed system would have a different purpose.

BizPortal focuses on providing access to official business services, whereas Business Buzz focuses on helping inexperienced entrepreneurs understand what they need to prepare and what steps they may need to follow before using those official services.

1.4.3 SARS eFiling

Platform: Web

SARS eFiling is a secure online environment through which taxpayers and businesses interact electronically with SARS. The service supports functions including returns and declarations, payments, tax-status viewing, and other tax-related interactions .

Figure 1.3 demonstrates the SARS eFiling environment and illustrates the use of secure user access and structured tax-service functions.

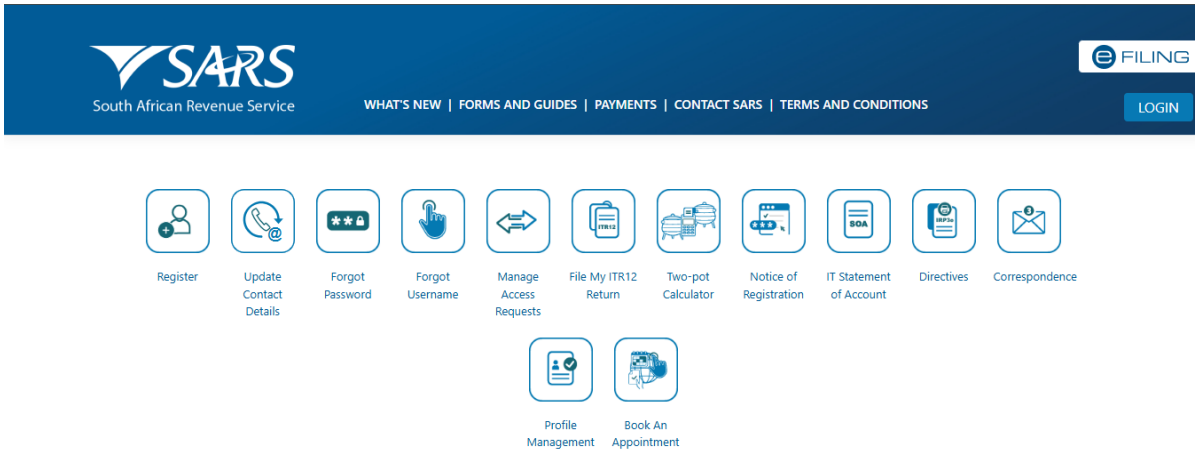


Figure 1.3: SARS eFiling interface Source: South African Revenue Service (SARS), 2025.

Features/processes to adapt:

- Secure user access.
- Compliance and status information.
- Electronic supporting-document handling.
- Record and history management.
- Clear separation of different tax-related functions.

Features/processes to avoid or improve:

- Do not replace SARS eFiling or submit real tax returns.
- Do not process real tax payments.
- Do not present simulated tax decisions as official SARS decisions.

- Use administrator-controlled source information for regulatory guidance displayed by the prototype.

Relevance to Business Buzz

SARS eFiling is relevant because tax administration is an important part of the business compliance journey. SARS itself explains that businesses may need to deal with different tax obligations depending on their circumstances.

Business Buzz would therefore help users understand which tax-related tasks may be relevant to their business and what they should prepare, while directing them to SARS for the actual tax transactions.

1.4.4 Comparative Analysis

Capability	**CIPC eServices**	**BizPortal**	**SARS eFiling**	**Business Buzz**
Business/company registration	Official	Official	Not primary	Simulated CIPC workflow
Tax-related services	Related	Related	Official	Simulated SARS workflow
UIF process	Not primary	Available	Not primary	Simulated when applicable
Identity verification	Related	Related	Not primary	Simulated Home Affairs stage
Centralized multi-service journey	Limited	Yes	No	Yes
Approval workflow	Institution-specific	Institution-specific	Institution-specific	Central Approver role
Document generation	Service-specific	Service-specific	Service-specific	Core feature
Print-ready hardcopy output	Process-specific	Process-specific	Process-specific	Core feature
Bank compliance review	Related bank services	Bank services available	No	Simulated Bank role

Real government transaction	Yes	Yes	Yes	No – simulation

The comparison shows that *Business Buzz* is not intended to compete with the official functions of CIPC or SARS. Its distinguishing feature is the coordination of selected institutional processes, documents, and approvals in a single simulated entrepreneur journey. The system also extends the centralization concept by including a dedicated Approver role and a Bank role that can review simulated compliance information. The Bank role is included to demonstrate how a centralized compliance view can support entrepreneurs in accessing financial services—a key barrier for SMMEs seeking funding.

1.4.5 Features and Processes to Adopt

- Secure authentication and role-based access.
- Centralized presentation of related services.
- Structured workflows and progress/status information.
- Business-profile information reused across multiple stages.
- Electronic document handling.
- Clear separation between official institutional services and simulated project functionality.

1.4.6 Features and Processes to Avoid

- Claiming that Business Buzz is an official CIPC, SARS, UIF, Home Affairs, or bank system.
- Claiming a formal partnership with an institution without documented authorization.
- Making real government or bank decisions.
- Depending on external APIs for core functionality when those APIs are not formally available to the project.
- Collecting unnecessary personal information that is not required for the prototype.

1.4.7 Identified Opportunity for Business Buzz

The related-system analysis identifies an opportunity to demonstrate a centralized workflow that coordinates selected business-registration and compliance stages rather than reproducing individual official platforms. Business Buzz will use a single business profile, conditional workflow logic, system-generated documents, approval stages and simulated institutional status updates to demonstrate this approach.

1.5 The project plan

A project plan is a document that contains the project scope and objectives. It answers key questions about what will be delivered, how it will be delivered on time, who is working on the project, and what milestones are set. This section presents the Business Buzz project plan, including the project information, scope, roles, stakeholders, communication approach, milestones, dependencies, and Gantt chart.

1.5.1 Project Scope and Deliverables

The project will deliver a functional web-based prototype of Business Buzz. The minimum viable product (MVP) will demonstrate the central entrepreneur journey and the interaction between the three application roles and the simulated institutional entities.

- User registration, login and role-based access for Entrepreneur, Approver and Bank users.
- Entrepreneur and business profile creation.
- Structured business-registration and compliance workflow.
- Simulated Home Affairs identity-verification stage.
- Simulated CIPC company-registration stage.
- Simulated SARS tax-related registration/compliance stage.
- Simulated UIF stage where business circumstances make it applicable.
- Approval and rejection workflow managed by the Approver.
- Automatic generation of digital documents using captured business information.
- Print ready hardcopy versions of generated documents.
- Centralized application and compliance status tracking.
- Simulated real-time status updates following workflow actions.
- Bank dashboard for reviewing relevant simulated business and compliance information.
- Simulated loan/financial-service assessment outcome.
- Audit trail for important workflow actions.
- Functional, usability, and security testing.
- Technical and user documentation and final system presentation.

1.5.2 Project Roles

Team Member	Role	Key Responsibilities

Anathi Silangwe - 227565649	Frontend developer and system analyst	Requirements analysis, process modelling, documentation, testing, and stakeholder feedback.
Talita Paca - 228141400	Project Manager and Documentation	Project coordination, technical planning, integration, version control, documentation, and delivery tracking.
Mnoneleli Nomboyo - 2234447311	Development / Database Lead	Database design, backend development, integration and technical testing.

The role allocation establishes accountability but does not restrict contribution. All team members will contribute to requirements, development, testing, documentation, and presentation.

1.5.3 Stakeholders and Governance

Stakeholder	Interest	Engagement
Entrepreneurs / small-business owners	Primary users need a clear and manageable business journey.	Requirements feedback, usability testing, and demonstration scenarios.
Approver	Reviews stages require human approval within the simulation.	Workflow validation and approval/rejection testing.
Bank role	Reviews relevant simulated business/compliance information.	Validation of compliance-view and simulated assessment workflow.
Academic lecturer / supervisor	Academic quality, scope and assessment compliance.	Scheduled reviews, feedback and final academic assessment/sign-off.
Project team	Successful delivery of a working system.	Weekly meetings, GitHub tasks/issues, peer review and testing.

Official institutional sources	Provide authoritative information used to model selected processes.	Source verification and careful labelling; no impersonation of official services.

1.5.4 How Will We Deliver on Time?

The project will be delivered on time using the following approach:

Approach	Description
Agile development	Work will be divided into phases with regular reviews and feedback
Priority-based development	Core MVP features will be implemented first; optional features added only after the core workflow is stable
Weekly progress meetings	Team members will meet weekly to track progress and address issues
GitHub version control	All code changes will be tracked, and separate branches will be used for parallel development
Early problem identification	Risks will be monitored and addressed early rather than left until final testing
Buffer time	Time has been allocated for unexpected delays and problem-solving
Regular lecturer consultations	Scheduled checkpoints will ensure the project remains aligned with expectations

1.5.5 Client Expectations, Feedback, Communication and Sign-off

Expectation	How It Will Be Met
A functional web-based prototype	The system will demonstrate the complete entrepreneur journey

Role-based access for three user types	Entrepreneur, Approver, and Bank environments will be implemented
Simulated institutional processes	CIPC, SARS, UIF, and Home Affairs processes will be simulated
Document generation capability	Documents will be generated automatically with watermarks
Compliance with academic requirements	The system will meet the scope defined in the approved proposal
Working increments throughout	Regular demonstrations will show progress

1.5. 6 Milestones, Dependencies and Gantt Plan

The schedule below is aligned with the supplied Semester 2 Teaching and Learning Plan and the professional Gantt chart prepared for Business Buzz. The final documents/system submission and system presentation remain TBC because the supplied academic schedule does not specify their exact dates.

Activity	**Start**	**End**	**Dependency**	**Deliverable / Milestone**
Project planning & problem definition	13 Jul 2026	24 Jul 2026	Project start	Problem definition and project direction
Business Case research & related-system analysis	27 Jul 2026	14 Aug 2026	Project direction	Research, literature and related systems
Business Case submission	17 Aug 2026	21 Aug 2026	Business Case completed	Assessment submission
Detailed requirements analysis	17 Aug 2026	04 Sep 2026	Business problem and feedback	Requirements backlog and process definition

Recess/development pause	27 Aug 2026	02 Sep 2026	Academic calendar	Scheduled recess
System requirements, specifications & technical design	03 Sep 2026	11 Sep 2026	Requirements analysis	Assessment submission
System design & database implementation	14 Sep 2026	25 Sep 2026	Technical design	ERD, schema, architecture, and base implementation
Implementation development	14 Sep 2026	02 Oct 2026	Design and database	Core system development
Implementation documentation submission	28 Sep 2026	02 Oct 2026	Implementation progress	Assessment submission
System integration & feature completion	20 Sept 2026	09 Oct 2026	Core components	Integrated workflow
Final consultation & refinement	4 Oct 2026	5 Oct 2026	Integrated system	Lecturer consultation and refinements
Final testing, bug fixing & security review	5 Oct 2026	16 Oct 2026	Integrated system	Testing and defect resolution
Final documentation & presentation preparation	10 Oct 2026	16 Oct 2026	Testing/refinement	Final report, demo data and presentation
Final documents & system submission	5 Oct 2026	16 Oct 2026	Final documentation	Final document submission
System presentation	21 Oct 2026	21 Oct 2026	Final system and presentation	Presentation date

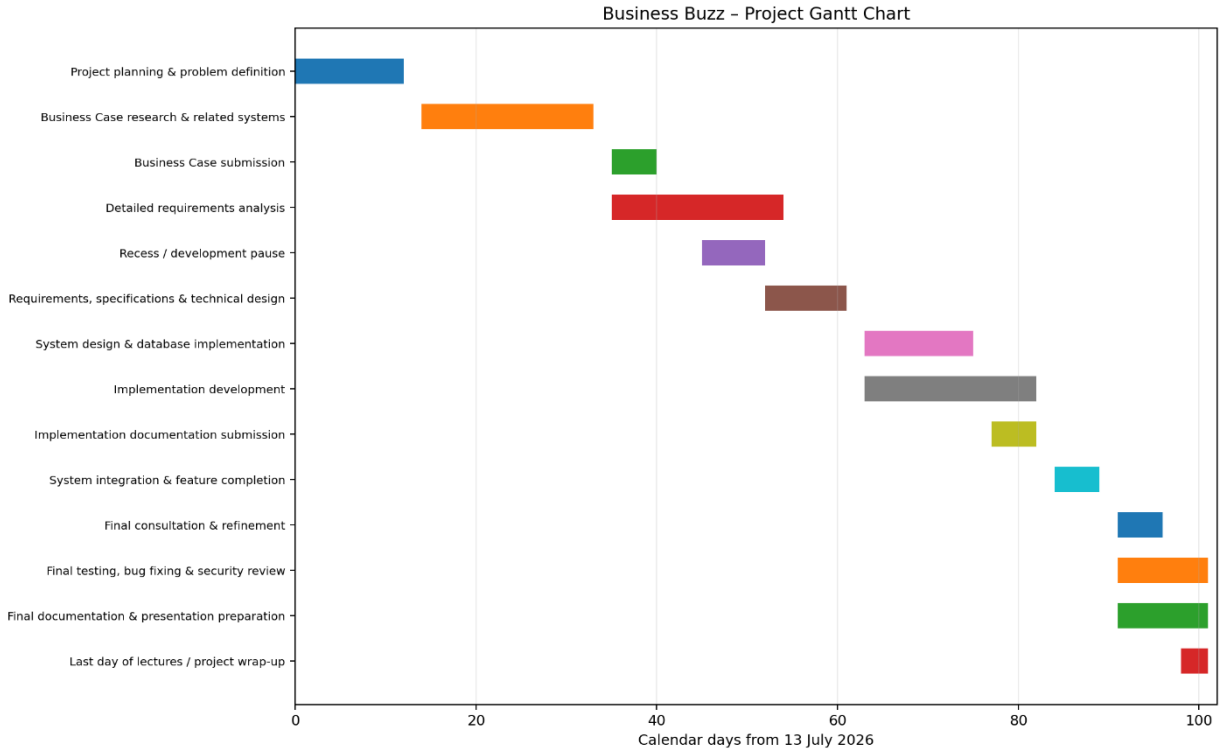


Figure 1.4: Business Buzz project Gantt chart.

Source: Project team schedule based on the supplied Semester 2 Teaching and Learning Plan (2026). Final submission and presentation dates are TBC.

1.5.7 Technical Delivery Approach

The current project direction requires a web-based system with a centralized workflow, role-based access, and simulated institutional entities. The implementation will use a relational database to store users, roles, businesses, workflow stages, approvals, documents, compliance statuses and audit records. The team will use GitHub for version control and collaborative development.

The system architecture will separate the user-facing application from core business logic and the database. Institutional simulations will be implemented as controlled internal modules/services rather than as claims of live government integration. This reduces dependency on external APIs and allows the team to demonstrate the complete workflow using controlled test data.

The implementation will prioritize the core MVP workflow: entrepreneur registration and business profile, institutional simulation stages, Approver review, document generation, compliance status and Bank review. Optional features will only be added after the core workflow is stable.

1.5.8 What Tools and Methods of Communication Will the Team Use?

GitHub: Version control, issue tracking, pull requests, code review, test documentation

Microsoft Teams: Team meetings, file sharing, instant messaging

WhatsApp: Quick communication and coordination

Email: Formal communication with lecturer

Visual Studio 2022/ Visual Code: Collaborative development, debugging, and testing

1.5.9 What Can Prevent the Project from Being a Success?

Risk	Mitigation
Scope becomes too large	MVP defined early; team agreement required before adding features
Incorrect representation of institutional processes	Use current official sources; label all outputs as simulated
Bugs discovered late in development	Continuous testing ensures bugs are identified and fixed early
Team member becomes unavailable	Document work; cross-review code; ensure more than one member understands each component
Loss or corruption of source code	Use GitHub; frequent commits; protected backups
Database changes break integration	Agree on ERD before major development; use versioned migrations
Time constraints	Prioritise MVP; buffer time allocated; regular progress tracking

1.6 Risk Analysis

Risk management will focus on events that could negatively affect project scope, quality, schedule, security, or the correctness of the simulated institutional workflow. The risk categories follow the Business Case template: acceptable, avoidable, minimizable, and transferable.

Risk	**Category**	**Likelihood**	**Impact**	**Treatment / Mitigation**

Scope becomes too large	Avoidable	High	High	Define an MVP early; priorities the Entrepreneur, Approver and Bank workflow; require team agreement before adding features.
Incorrect representation of CIPC/SARS/UIF/Home Affairs processes	Minimizable	Medium	High	Use current official sources; record source dates; model only selected processes; label all outputs as simulated.
Dependence on external government or bank APIs	Acceptable	Low	High	Do not make core functionality dependent on live APIs. Use controlled internal simulations unless an authorized integration becomes available.
Document generation produces incorrect or incomplete information	Minimizable	Medium	High	Use validated templates, required-field checks, test data and document review before release.
Role-authorisation failure exposes information	Minimizable	Medium	High	Implement role-based access control, server-side authorisation, input validation and security testing.
Approval workflow behaves incorrectly	Minimizable	Medium	High	Define workflow states formally, test approve/reject/request-information paths and maintain an audit trail.

Database changes break integration	Minimizable	Medium	Medium	Agree on the ERD before major development, use versioned migrations/scripts and test changes before merging.
Team member becomes unavailable	Minimizable	Medium	High	Document work, use GitHub, cross-review code and ensure more than one member understands each major component.
Loss or corruption of source code	Avoidable	Low	High	Use GitHub, frequent commits, branches and protected backups.
Final submission or presentation is incomplete	Minimizable	Medium	High	Maintain a submission checklist, complete documentation alongside development and schedule a final rehearsal.

1.6.1 Risk Priorities

The highest priority risks are scope expansion, incorrect institutional modelling, security/ role-authorization failures, and document/approval workflow errors. These risks directly affect whether the final demonstration will be credible and functional. The team will address them early rather than leaving them until the final testing period.

2. System Requirements
Specifications and Technical Design

This document presents the System Requirements Specifications and Technical Design for Business Buzz, a proposed centralized business registration and compliance management system. The purpose of this document is to translate the objectives identified in the Business Buzz Business Case into clearly defined system requirements and a planned technical design that will guide the development of the system.

Business Buzz is intended to provide entrepreneurs with a centralized digital environment through which selected business registration and compliance activities can be completed and monitored. The system will simulate selected processes associated with the Companies and Intellectual Property Commission (CIPC), South African Revenue Service (SARS), Unemployment Insurance Fund (UIF), and Department of Home Affairs. These institutional processes are represented within the project for academic and demonstration purposes and do not constitute direct integration with, or official services provided by, these institutions.

The system will support three primary user roles: Entrepreneur, Approver, and Bank. Entrepreneurs will be able to create and manage business profiles, progress through simulated registration and compliance stages, generate relevant documents, and monitor the status of their applications. Approvers will review submitted applications and make approval or rejection decisions where human review is required. Bank users will be able to review relevant business registration and compliance information and perform a simulated financial-service or loan assessment.

This document defines the planned solution for addressing the project's objectives, the information, functional and communication scope of the system, stakeholder requirements, hardware and software requirements, and relevant design constraints. It further presents the planned high-level use case diagram, UML class model, relational database model and user interface designs. Together, these specifications will provide a foundation for the implementation, testing and evaluation of Business Buzz.

2.1 Plan to Address Objectives

The Business Buzz Business Case identified the need for a more centralized and understandable approach to selected business registration and compliance processes. Business Buzz will address these objectives by providing a single web-based platform through which entrepreneurs can capture business information, progress through simulated institutional processes, monitor application and compliance status, and access system-generated documentation.

2.1.1 Centralize Selected Business Registration and Compliance Processes

Business Buzz will provide a centralised web-based platform that combines selected simulated processes associated with Home Affairs, CIPC, SARS and UIF. Instead of representing these activities as completely separate systems, Business Buzz will organise

them into a coordinated workflow accessible through a single authenticated user account. An entrepreneur will create a profile and capture relevant personal and business information. Where appropriate, this information will be reused across subsequent stages of the workflow. The entrepreneur will therefore have a central location from which to initiate activities, identify outstanding requirements, and monitor overall progress.

2.1.2 Simplify the Registration and Compliance Process

Business Buzz will provide a structured, step-by-step workflow that guides entrepreneurs through the applicable stages of the simulated registration and compliance process. The interface will indicate the current stage, completed stages and outstanding actions. Forms, validation rules and clear instructions will assist entrepreneurs when capturing information. Required fields will be identified and information validated before applicable submissions proceed. Where a process is conditional, such as UIF-related activities, relevant business information will be used to determine whether the stage applies.

2.1.3 Reduce Duplication of Information

Entrepreneur and business information will be maintained within a central relational database. Information that has already been captured and remains applicable to another stage will be retrieved from the database rather than requiring the entrepreneur to repeatedly enter the same information. For example, information stored in entrepreneur and business profiles may be reused when preparing registration, compliance and system-generated documentation. This will reduce repetitive data entry and improve consistency between different parts of the system.

2.1.4 Provide Transparent Status Tracking

Business Buzz will provide a dashboard through which entrepreneurs can monitor their registration and compliance activities. Applicable stages will have identifiable statuses such as Not Started, In Progress, Pending Review, Approved, Rejected, or Completed, depending on the process. Where an application requires human review, it will be made available to an authorized Approver. The resulting decision will be reflected within Business Buzz so that the entrepreneur can determine the outcome and any further action required. Status information displayed by Business Buzz represents activity within the prototype and does not represent live status information obtained from government institutions.

2.1.5 Generate Relevant Business Documents

Business Buzz will use information stored in entrepreneur and business profiles to generate relevant digital documents. Generated documents will be available for electronic viewing and, where applicable, as print-ready hardcopy documents. Using information already captured and validated by the system will improve consistency and reduce unnecessary document preparation. Documents produced by Business Buzz will be presented as prototype-generated documents and will not be represented as official documents issued by CIPC, SARS, UIF, Home Affairs or any other institution.

2.1.6 Support Controlled Application Review

Business Buzz will provide an Approver role for stages requiring simulated human assessment. Authorized Approvers will be able to review relevant application details and supporting information and record an approval, rejection or request for additional information where applicable. Decisions and associated comments will be stored against the relevant application, providing a record of the review process. Role-based access control will restrict this functionality to authorized users.

2.1.7 Support Compliance Visibility and Financial Assessment

Business Buzz will provide a consolidated view of relevant simulated business registration and compliance information. Once the applicable stages have been completed, entrepreneurs will be able to view the current registration and compliance position of their business. The Bank role will allow authorized Bank users to review relevant business registration and compliance information and record the outcome of a simulated financial-service or loan assessment. This functionality is for academic demonstration purposes and will not perform actual credit checks, issue loans or represent decisions made by a real financial institution.

2.1.8 Improve Accountability and Traceability

Business Buzz will maintain records of significant activities performed within the system. Relevant events may include application submissions, status changes, approval or rejection decisions, document generation and financial assessments. Where applicable, these activities will be associated with the responsible user and date and time, creating an audit history that supports accountability and traceability.

2.2 Project scope

The scope of Business Buzz defines the information handled by the system, the functionality provided to users and the manner in which users and system components communicate.

Business Buzz will be developed as a web-based academic prototype supporting three primary user roles: Entrepreneur, Approver and Bank. Selected processes associated with Home Affairs, CIPC, SARS and UIF will be simulated to demonstrate how business registration and compliance activities could be coordinated through a centralised system.

The project does not include direct integration with official government databases, banking systems or institutional APIs. Identity verification, registration, compliance processing and financial assessments performed within Business Buzz will therefore be simulated for academic and demonstration purposes.

2.2.1 Information Scope

Business Buzz will capture, process, store and retrieve information required to support the simulated business registration and compliance workflow. Information will be stored in a central relational database and associated with relevant users, businesses, applications and workflow activities.

User and Entrepreneur Information

The system will maintain information required to identify and authenticate users, including names, contact information, login information, account status and assigned role.
Entrepreneur profiles will contain the personal and contact information required by the Business Buzz workflow. This may include names, identification information, contact details and address information.
Authentication credentials will be handled securely, and passwords will not be stored as readable plain text.

Business Information

Business profiles will contain information required throughout the registration and compliance workflow. This may include the business name, business type, industry or business activity, business address, contact information, registration information, ownership or director information and relevant employment information.
Information already available within the business profile will be reused by applicable workflow stages where possible.

Application and Workflow Information

The system will maintain information relating to applications and their progression through Business Buzz. This may include application reference numbers, submission dates, workflow stages, statuses, approval or rejection decisions, decision dates, comments and completion information.
This information will allow entrepreneurs to monitor progress and Approvers to identify applications requiring review.

Identity-Verification Information

Business Buzz will record the outcome of its simulated Home Affairs identity-verification stage. A result may indicate that verification is pending, successful or unsuccessful.
The prototype will not query actual Department of Home Affairs systems or databases.

Registration and Compliance Information

The system will maintain information relating to simulated CIPC, SARS and UIF stages, including relevant statuses, dates, references and stage outcomes.
UIF-related information will only be processed where the circumstances represented in the system make the stage applicable.
These records will represent the state of processes within Business Buzz rather than official institutional records.

Document Information

Business Buzz will maintain records of documents generated during the registration and compliance workflow. Document information may include document type, associated business or application, generation date, status and retrieval information.
Generated documents will be available electronically and, where applicable, in print-ready form.

Bank Assessment Information

Business Buzz will maintain information associated with simulated financial assessments. This may include the business assessed, assessment date, status, outcome and associated comments.
The information will be used solely for demonstration and will not constitute an actual banking or credit decision.

Audit Information

The system will maintain audit information for significant activities. This may include the user responsible for an action, the action performed, the affected record and the date and time of the activity.

2.2.2 Functional Scope

The functional scope defines the functionality Business Buzz is expected to provide.

User Registration and Authentication

Entrepreneurs will be able to create accounts and authenticate themselves before accessing protected functionality. Approver and Bank accounts will be assigned appropriate permissions.
Role-based access control will determine the information and functionality available to each user.

Profile Management

Entrepreneurs will be able to create and maintain personal and business profiles. Information already available within these profiles will be reused where applicable to reduce unnecessary duplication.

Simulated Identity Verification
Business Buzz will provide a simulated Home Affairs identity-verification process. Information captured by the entrepreneur will be evaluated using rules or predetermined prototype data to produce a simulated result.

Simulated Business Registration

The system will provide functionality representing selected aspects of CIPC-related business registration. Entrepreneurs will capture required business information and submit applications through Business Buzz.

Where required, applications will be routed to an Approver for review. Any registration-related output generated by the system will be clearly presented as prototype output rather than an official CIPC registration.

Application Review
Authorized Approvers will have access to applications requiring review. They will be able to inspect relevant information and record an approval, rejection or request for additional information.
The system will record the decision and update the relevant application status.

Simulated Tax and UIF Compliance
Business Buzz will provide simulated SARS-related functionality for applicable businesses and maintain the resulting status.

Where relevant business circumstances require it, the system will also provide a simulated UIF-related stage. Businesses for which UIF is not applicable will not be required to complete the stage.

Application and Compliance Tracking
Entrepreneurs will have access to a dashboard displaying the current state of registration and compliance activities. It will identify completed stages, pending reviews, unsuccessful stages and outstanding actions.

Document Generation
Business Buzz will generate selected documents using information captured within the system. Users will be able to view applicable documents electronically and obtain print-ready versions where required.

Notifications
Business Buzz will provide in-system notifications for significant events, including application submissions, approvals, rejections, requests for additional information and relevant status changes.

Bank Review and Simulated Financial Assessment
Authorized Bank users will be able to review businesses submitted for simulated financial assessment. Relevant registration and compliance information and permitted documents will be made available for review.

Bank users will be able to record a simulated assessment outcome. Business Buzz will not perform actual credit-bureau checks, calculate official credit scores, transfer funds or issue real loans.

Audit Trail

The system will record selected significant actions, including application submissions, approval decisions, status changes, document generation and Bank assessment outcomes.

Functional Boundaries
Business Buzz will not provide direct access to or integration with live CIPC, SARS, UIF or Home Affairs databases. It will not perform official government submissions, actual Home Affairs identity verification, real banking integration, actual credit checks or real loan approvals.

The system will not represent itself as an authorized replacement for the official systems of the institutions being simulated.

1.2.3 Communication Scope

Users will interact with Business Buzz through a responsive web-based interface. Entrepreneurs, Approvers and Bank users will authenticate before accessing functionality associated with their respective roles.

User requests will be processed by the application's backend, which will apply the relevant business rules and retrieve or store information in the central database.

Business Buzz will use in-system notifications to communicate significant workflow events. Notifications may be generated when an application is submitted, approved or rejected, additional information is required, a registration or compliance stage changes, an application requires Approver review, a document becomes available or a business is submitted for Bank assessment.

Communication between user roles will occur primarily through structured system actions rather than unrestricted direct messaging. For example, an Approver may provide a reason for rejecting an application or requesting additional information, which will then be made available to the Entrepreneur through the relevant application.

The Business Buzz web interface will communicate with the ASP.NET Core Web API, while the API will perform authorized database operations against Microsoft SQL Server. Ordinary users will not be permitted to communicate directly with the database.

Business Buzz will require an internet or suitable network connection. Offline processing and offline synchronization are outside the current project scope.

The prototype will not communicate directly with live systems operated by CIPC, SARS, UIF, Home Affairs or financial institutions. Relevant institutional interactions will be simulated internally.

2.3 Business requirements

Business requirements, also referred to as stakeholder requirements, describe the characteristics of Business Buzz from the perspective of its end users. The system has three

primary end-user types: Entrepreneur, Approver and Bank. Each has different responsibilities and therefore requires different functionality and levels of access.

2.3.1 Entrepreneur Requirements

Entrepreneurs require an accessible and understandable environment through which they can manage business information and progress through the applicable simulated registration and compliance stages.
Entrepreneurs require the ability to:

- create an account and securely authenticate;
- create and maintain a personal profile;
- create and manage a business profile;
- capture ownership or director information where applicable;
- progress through a structured registration and compliance workflow;
- complete the simulated identity-verification stage;
- review information before submitting an application;
- submit applications and receive an application reference;
- monitor application and compliance statuses;
- view reasons for rejection or requests for additional information;
- access and print applicable generated documents; and
- receive in-system notifications regarding significant workflow events.

The Entrepreneur interface should clearly indicate completed stages, current activities and outstanding actions. Information already available within Business Buzz should be reused where applicable to minimize repetitive data entry.

2.3.2 Approver Requirements

Approvers require secure and controlled access to applications requiring human review.
Approvers require the ability to:

- securely authenticate using an authorized Approver account;
- access a dashboard containing applications requiring review;
- identify applications by relevant information such as application reference, business, submission date and status;
- review relevant entrepreneur and business information;
- review applicable generated documents;

- approve an application;
- reject an application;
- request additional or corrected information where appropriate;
- provide comments or reasons associated with a decision;
- access relevant review history; and
- receive notification of applications requiring attention.

Following an Approver's decision, Business Buzz should update the relevant application status and make the result available to the Entrepreneur.
The system should record the responsible Approver and relevant decision information to support traceability.

2.3.3 Bank Requirements

Bank users require controlled access to relevant information concerning businesses submitted for simulated financial assessment.
Bank users require the ability to:

- securely authenticate using an authorized Bank account;
- access a dashboard of businesses submitted for assessment;
- review permitted business profile information;
- review relevant simulated registration and compliance statuses;
- access permitted documents;
- record the result of a simulated financial assessment;
- provide relevant comments or reasons; and
- access applicable assessment history.

Bank users must not have unrestricted access to information unrelated to their assessment responsibilities.
The financial-assessment functionality will be used solely for academic demonstration and will not constitute an actual credit assessment, credit score, loan approval or financial transaction.

2.3.4 Common User Requirements

All Business Buzz users require a system that is secure, reliable, responsive and easy to navigate.
The system should provide consistent navigation and presentation, clear feedback after important actions and understandable validation or error messages.

Users must only be able to access information and functionality permitted by their assigned role. Significant actions that affect application or assessment outcomes should also require appropriate confirmation before being finalized.

2.4 Hardware and Software Requirements

Business Buzz will require software development tools, database technologies, web technologies and appropriate computing hardware. The technology stack will remain consistent with that defined in the Business Buzz Project Proposal and Business Case.

2.4.1 Software Requirements

Microsoft Visual Studio 2022
Microsoft Visual Studio 2022 will serve as the primary integrated development environment for Business Buzz. It will be used to create, manage, debug and test the application.

ASP.NET Core Web API
The Business Buzz backend will be developed using ASP.NET Core Web API. The API will process requests from the web interface, implement business rules and workflow logic, enforce authentication and authorisation, and communicate with the database.

C#
C# will be the primary server-side programming language. It will be used to implement business logic, validation, workflow processing and database-related functionality.

HTML5
HTML5 will define the structure and content of the Business Buzz web interfaces, including forms, dashboards, navigation components and tables.

CSS3
CSS3 will control the visual presentation of the system and provide consistent styling, spacing, typography and layout.

Bootstrap
Bootstrap will support the development of a responsive interface capable of adapting to desktop, laptop, tablet and smartphone screen sizes.

JavaScript
JavaScript will provide client-side interactivity, including applicable validation, dynamic interface behaviour, user feedback and communication with backend functionality.

Microsoft SQL Server
Microsoft SQL Server will serve as the relational database management system. It will store information relating to users, businesses, applications, workflows, approvals, compliance activities, documents, notifications, Bank assessments and audit activities.

The final relational database model will contain at least the required number of tables specified for a three-person project group. The lecturer's template specifies a minimum equivalent of 14 tables for groups of three students.

Git and GitHub
Git will be used for version control, enabling the development team to track source-code changes and collaborate through branches and merges.
GitHub will serve as the remote repository for the project, providing a central location for collaboration and maintaining the project's source-code history.

Web Browser
Business Buzz will require a modern web browser. Current versions of commonly available browsers will be supported, with Google Chrome serving as the primary browser for development and testing.

2.4.2 Hardware Requirements

Each developer will require access to a desktop or laptop capable of running Visual Studio 2022, Microsoft SQL Server, and the associated development tools.

Recommended development hardware includes a 64-bit processor, at least 8 GB of RAM, sufficient storage for the development environment and project files, and an internet connection for GitHub collaboration and other online development activities.

End users will not require specialized hardware. Entrepreneurs, Approvers, and Bank users will require a desktop computer, laptop, tablet, or smartphone capable of running a modern web browser and connecting to the internet or an appropriate network.

Business Buzz will also require a suitable server environment capable of hosting the ASP.NET Core application and supporting access to the Microsoft SQL Server database. During development and testing, the system may operate within an appropriate local or test environment, with final deployment dependent on the hosting resources available to the project team.

2.5 Design constraints

The development of Business Buzz will be influenced by security, interface and performance constraints. These constraints must be considered throughout development to ensure that the system remains secure, usable and capable of supporting its intended functionality.

2.5.1 Security Constraints
Business Buzz will process personal, business, application and compliance-related information. Appropriate controls must therefore be implemented to protect information against unauthorized access, modification or disclosure.

Authentication and Password Security

Users must authenticate before accessing protected areas of Business Buzz. Passwords must not be stored as readable plain text and should be handled using appropriate secure password-hashing and verification mechanisms.
Unauthenticated users must not have access to protected Entrepreneur, Approver or Bank functionality.

Role-Based Access Control
Business Buzz will implement role-based access control for Entrepreneur, Approver and Bank users.
Entrepreneurs should only access information associated with their own accounts and businesses. Approvers should access information required for application review, while Bank users should only access information required for simulated financial assessments.
Access controls must be enforced by the backend and should not depend solely on hiding interface elements.

Data Protection and Validation
Database access will occur through authorized backend functionality rather than direct access by ordinary users.
Information submitted by users must be validated for required fields, appropriate formats, expected data types and applicable business rules. Important validation and security rules must be enforced on the server side.

Secure Communication and Sessions
Where deployed over a network, communication between the browser and Business Buzz should use HTTPS where supported by the deployment environment.
Authenticated sessions should be securely managed, and users should not remain authenticated indefinitely. Logging out should terminate access through the authenticated session.

Document Security
Generated documents must only be accessible to authorized users. Users should not be able to access documents belonging to unrelated businesses by manipulating document identifiers or requests.

Audit Trail
Significant activities should be recorded to support accountability and traceability. These may include application submissions, approvals, rejections, requests for additional information, status changes, document generation and Bank assessment decisions.

Simulation Boundary
Business Buzz must clearly distinguish its simulated functionality from official institutional processes. It must not claim that prototype-generated statuses or documents represent official Home Affairs, CIPC, SARS or UIF outcomes or actual decisions made by a financial institution.

2.5.2 Interface Constraints

Business Buzz will be used by users with different levels of technical experience. The interface should therefore be responsive, consistent and straightforward to navigate.

The web interface will be developed using HTML5, CSS3, Bootstrap and JavaScript and should adapt appropriately to desktop, laptop, tablet and smartphone displays.

Navigation, buttons, forms, headings and status indicators should follow consistent patterns throughout the system. Navigation options should also reflect the authenticated user's role so that users are not presented with functionality they cannot access.

The registration and compliance workflow should clearly indicate the user's progress. Statuses such as Not Started, In Progress, Pending Review, Approved, Rejected, Additional Information Required, and Completed should be displayed consistently.

Forms should clearly identify required information and provide understandable validation messages. Where appropriate, dropdown lists, date controls and other suitable input components should be used to reduce incorrect data entry.

Significant actions, such as submitting an application or finalizing an approval, rejection or assessment decision, should provide appropriate confirmation before completion.

The interface should maintain readable text, appropriate spacing and clearly identifiable controls while avoiding unnecessary complexity.

2.5.3 Performance Constraints

Business Buzz should provide acceptable performance during normal academic prototype usage.

Frequently accessed pages, particularly the Entrepreneur, Approver and Bank dashboards, should retrieve and display information efficiently. Database queries should retrieve only the information necessary for the requested functionality.

Microsoft SQL Server should be structured using appropriate primary keys, foreign keys and relevant indexing considerations to support efficient retrieval of frequently accessed records.

Server-side functionality implemented through ASP.NET Core should avoid unnecessary processing. Operations such as application submission, status updates, approval decisions and Bank assessments should provide appropriate feedback after successful completion or when an error occurs.

Document generation should complete within a reasonable period under normal prototype usage and should reuse information already available within Business Buzz where appropriate.

Although the prototype is not expected to support the scale of a national government or commercial banking platform, multiple authorized users should be able to perform normal

system activities without one user's operations incorrectly affecting another user's information.

As a web-based system, Business Buzz will depend on network connectivity and the performance of its hosting environment. Offline functionality and offline synchronizations are not included in the current scope.

The application should also handle foreseeable errors without unnecessarily exposing technical information or causing the entire system to become unusable. Where an operation fails, users should receive an understandable message and be allowed to retry where appropriate.

2.6 High-level use case diagram

The high-level use case diagrams illustrate the main interactions between the users of the Business Buzz system and the functionality provided by the system. The diagrams are divided into four complementary views to improve clarity and ensure that the responsibilities of each role can be understood independently while still showing how the different roles contribute to the overall business journey.

The first view presents the overall Business Buzz system, showing how the main user roles interact with the system and how their activities contribute to the broader entrepreneur journey. The second view focuses specifically on the Entrepreneur, who uses Business Buzz to provide business information, receive guidance from BuzzGuide, prepare required information and documents, progress through applicable processes and monitor the status of their journey.

The third view focuses on the Approver, representing an authorized institutional reviewer within the Business Buzz system. This view shows how applications and submitted information can be reviewed, approved, rejected or returned to the entrepreneur when additional information is required. The fourth view focuses on the Bank User, showing how relevant business and compliance information can be reviewed as part of the system's simulated funding or financial-service assessment.

Together, these four views provide both role-specific detail and an overall representation of how the system components work together. The diagrams therefore avoid placing all functionality into a single diagram while still maintaining a consistent relationship between the Entrepreneur, Approver and Bank User within the Business Buzz workflow.

Overall system Use case Modelling

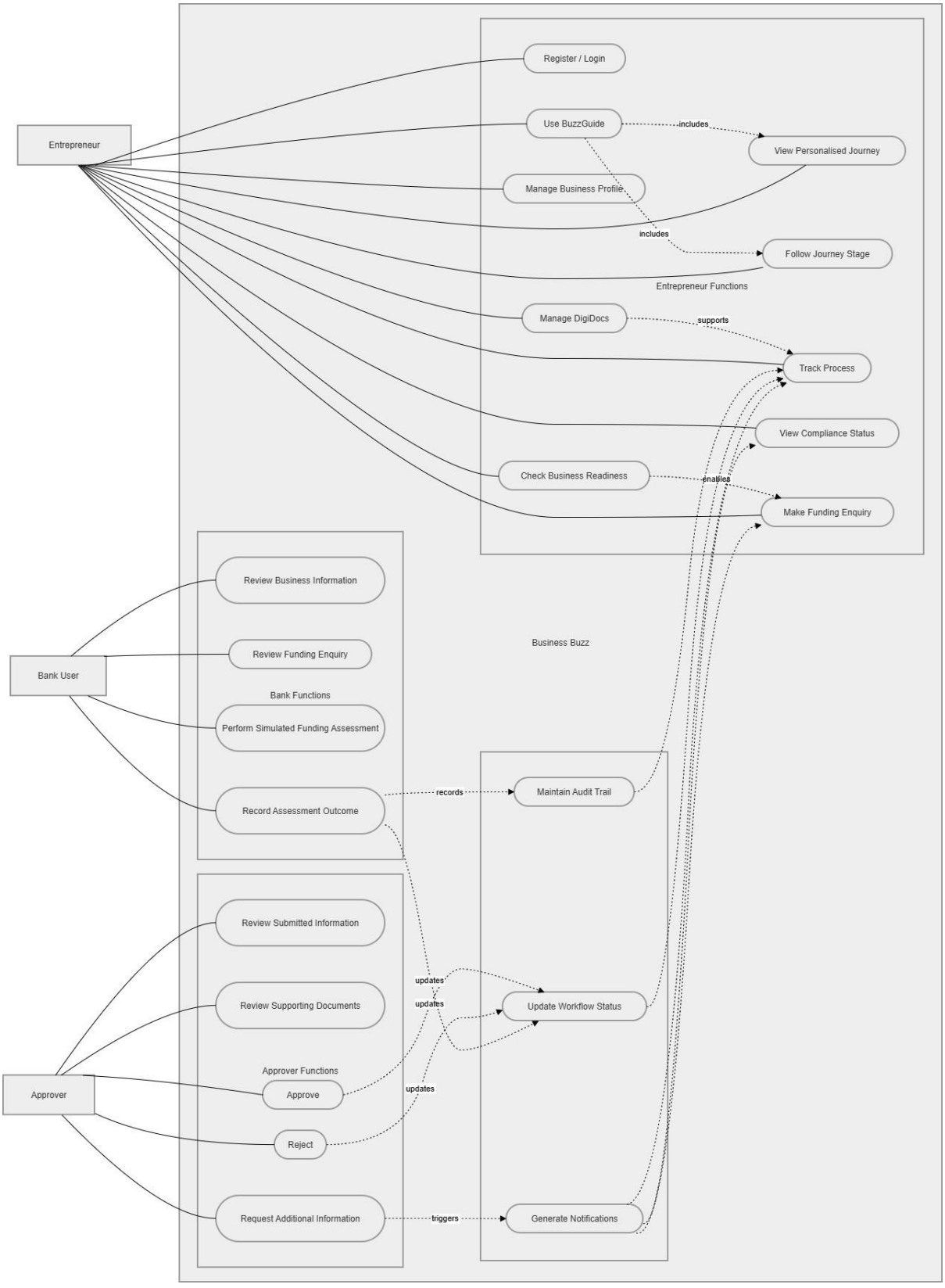


Figure 2.1: Overall system Use case Modelling

Entrepreneur Use Case Function Modelling

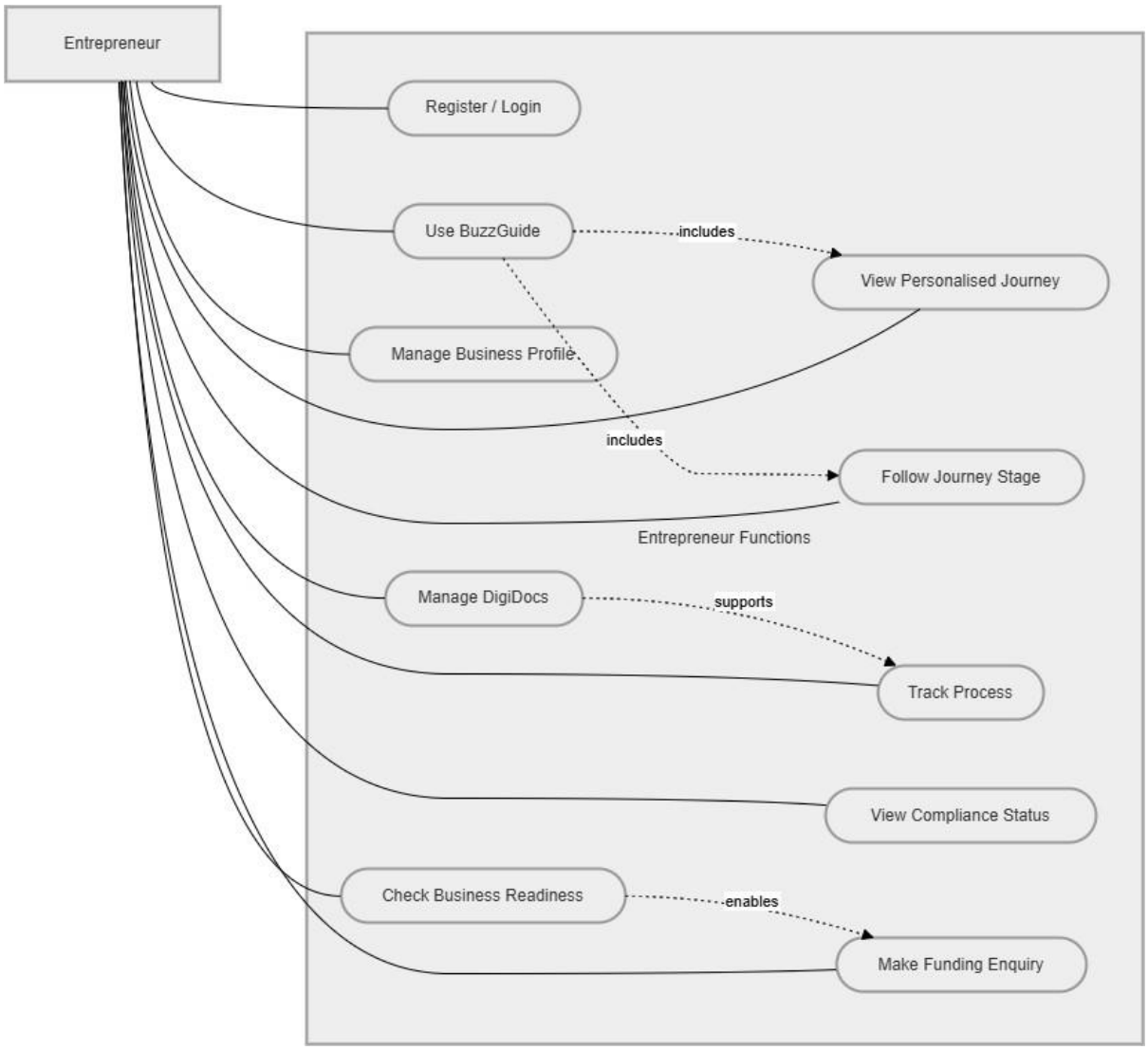


Figure 2.2: Entrepreneur Use Case

Bank Use Case Function Modelling

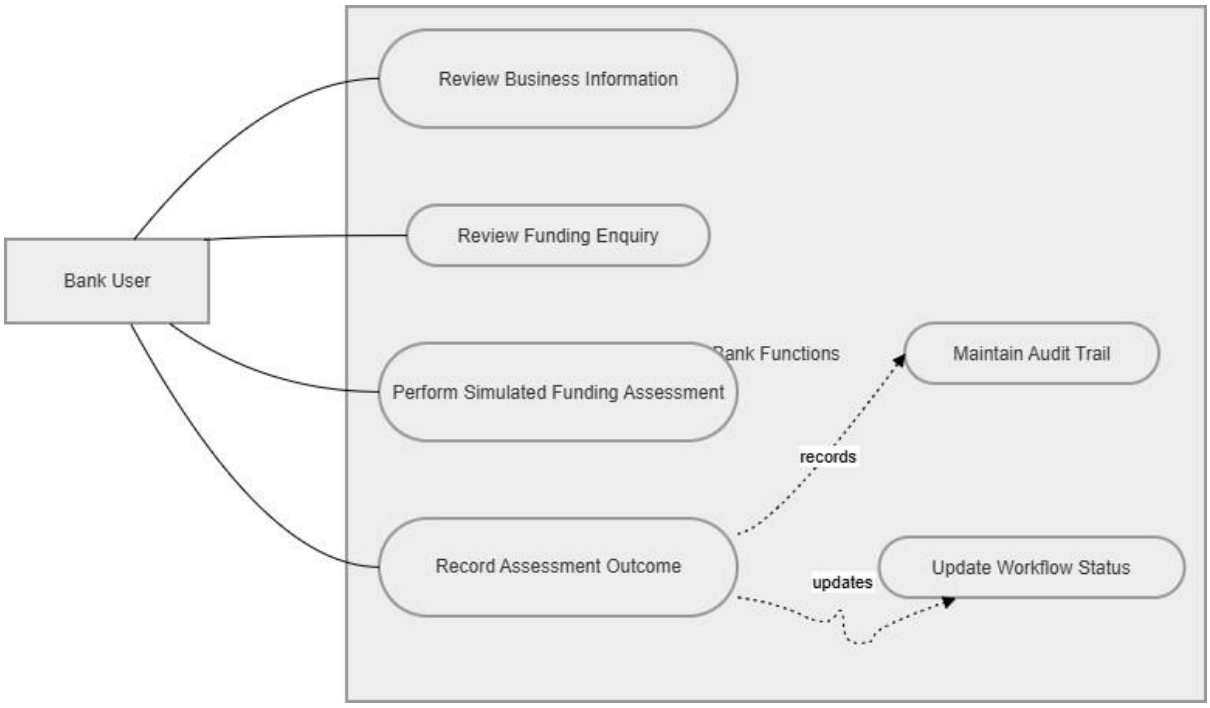


Figure 2.3: Bank Use Case

Approver Use Case Function Modelling

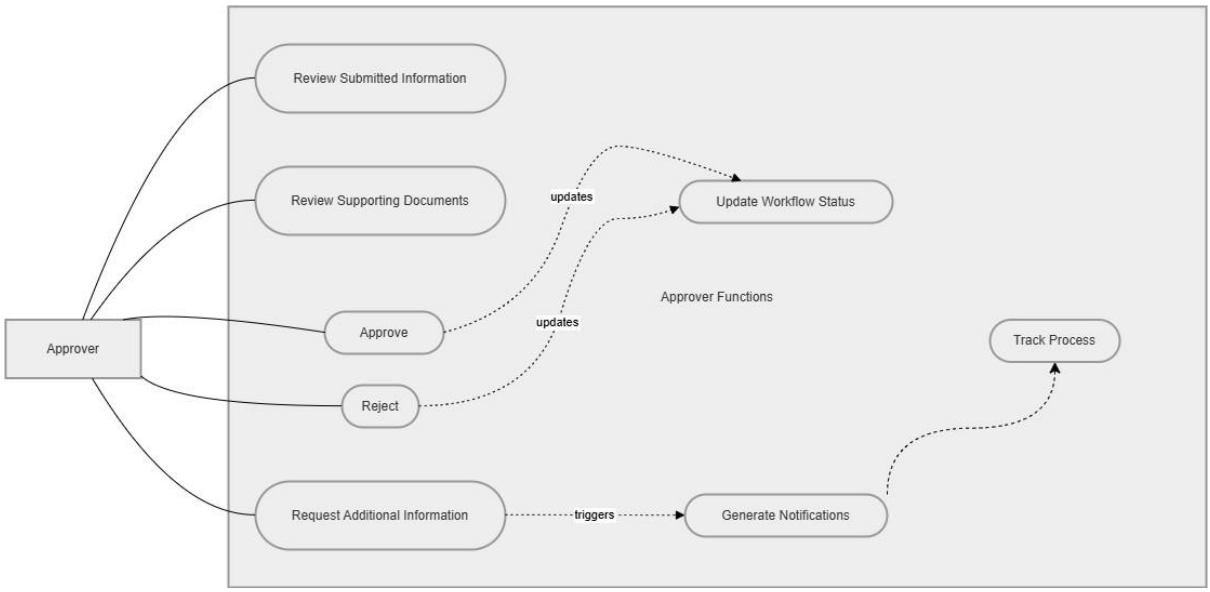


Figure 2.4: Approver Use Case

2.7 UML Class Model / Diagram

This outlines the UML class diagram designed for the Business Buzz system. Developed with a C# Entity Framework Core Code-First approach in mind, this architectural model serves as the structural blueprint for the backend database and application logic. The diagram consists

of fourteen core classes that comprehensively capture user authentication, role-based access control, business profile management, institutional journey stages, document handling, compliance monitoring, and automated auditing.

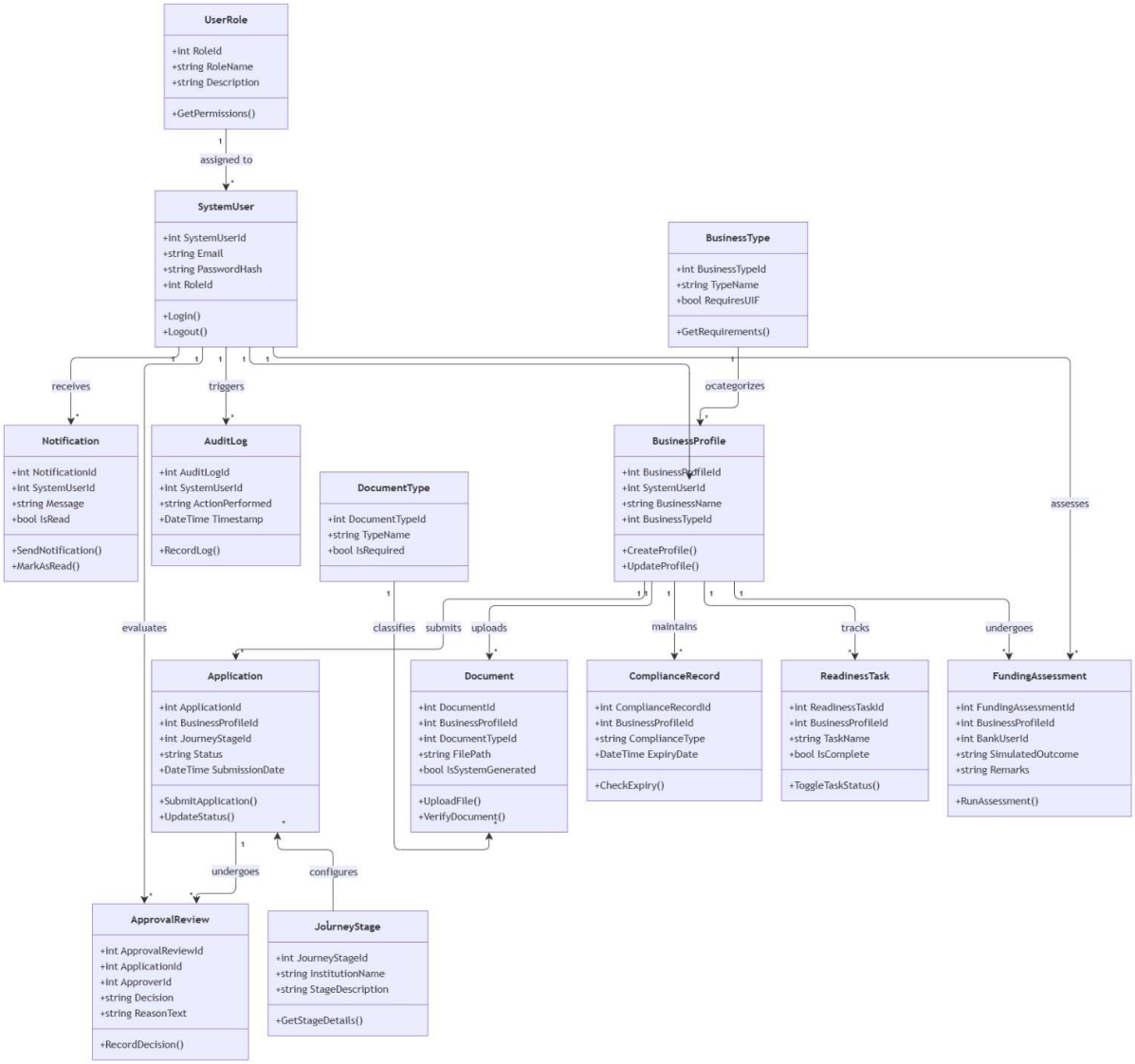


Figure 2.5: represents the UML class diagram of the Business Buzz system.

2.8 Relational Database Model Diagram

Figure 2.6 below presents the Entity Relationship Diagram (ERD) for the Business Buzz system. The diagram illustrates the planned relational database structure and shows how the different entities within the system are connected through primary and foreign key relationships.

The database is designed to support the main Business Buzz processes, including user and role management, entrepreneur and business information, business applications, application workflow and approvals, compliance processes, document management, notifications, bank

assessments, and audit activities. These entities work together to support the movement of an entrepreneur's business through the different registration, compliance, approval, and financial assessment stages of the system.

The relationships shown in the ERD are intended to maintain data integrity and traceability by ensuring that records such as applications, documents, approvals, compliance information, and assessments can be associated with the relevant user or business. The model also supports the system's role-based structure by separating the information and activities required by Entrepreneurs, Approvers, and Bank users.

Overall, the relational database model provides the data foundation required for Business Buzz to centrally store and manage information throughout the simulated business registration and compliance process.

Figure 2.6 represents the entity relationship diagram of the YOURPLANNEDSYSTEMNAME system.

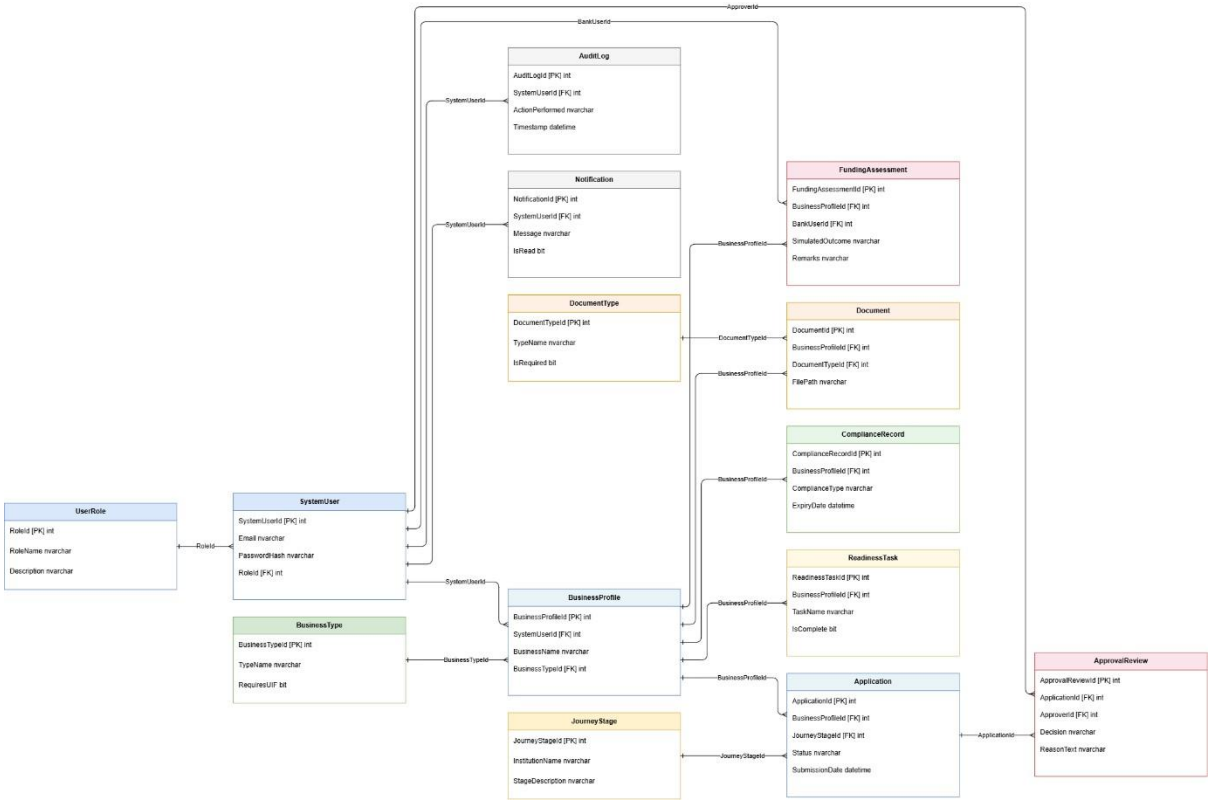


Figure 2.6: represents the UML class diagram of the Business Buzz system.

2.9 User Interface Design

The user interface design for Business Buzz illustrates the planned screens through which the different users will interact with the system. The interfaces have been designed to provide a clear, consistent and user-friendly experience while ensuring that functionality is presented according to the responsibilities and access permissions of each user role.

The interface is divided into three main environments: the Entrepreneur Interface, Approver Interface and Bank Interface. The Entrepreneur interface supports account access, business profile management, progress tracking, document management, compliance readiness and funding enquiries. The Approver interface provides functionality for reviewing submitted applications, monitoring applications requiring attention and managing requests for additional information. The Bank interface enables authorized Bank users to review business readiness information, manage funding enquiries and perform simulated financial-service assessments.

The interfaces follow a consistent Business Buzz visual design and use dashboards, navigation menus, status indicators, forms and action buttons to assist users in completing their respective tasks. The designs also clearly distinguish simulated functionality from official government or banking processes where applicable.
The following figures present the proposed Business Buzz user interfaces and demonstrate how each user type will interact with the main functionality of the system.

2.9.1 Entrepreneur Interface Design

Hero Start-up Page

Figure 2.7 presents the Business Buzz Hero Start-up Page. This screen introduces the Business Buzz platform to the entrepreneur and provides options to begin the business journey or learn more about how the system works. It serves as the main entry point into the system.

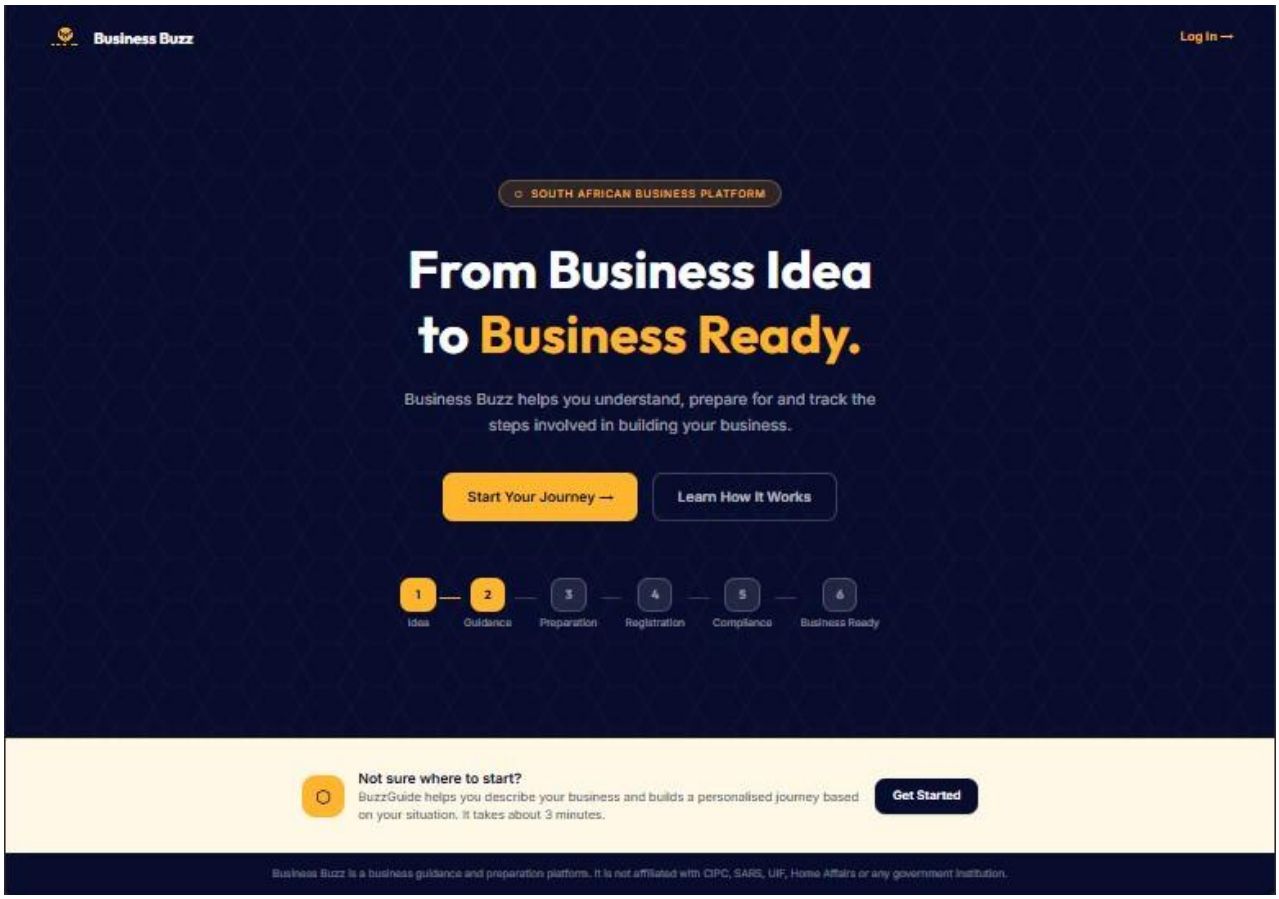


Figure 2.7: Hero Start-up Page

Login and Sign-Up Page

Figure 2.8 presents the Login and Sign-Up interface. New entrepreneurs can create an account by providing their personal login details, while existing users can sign in using their registered email address and password. Successful authentication provides access to the entrepreneur environment.

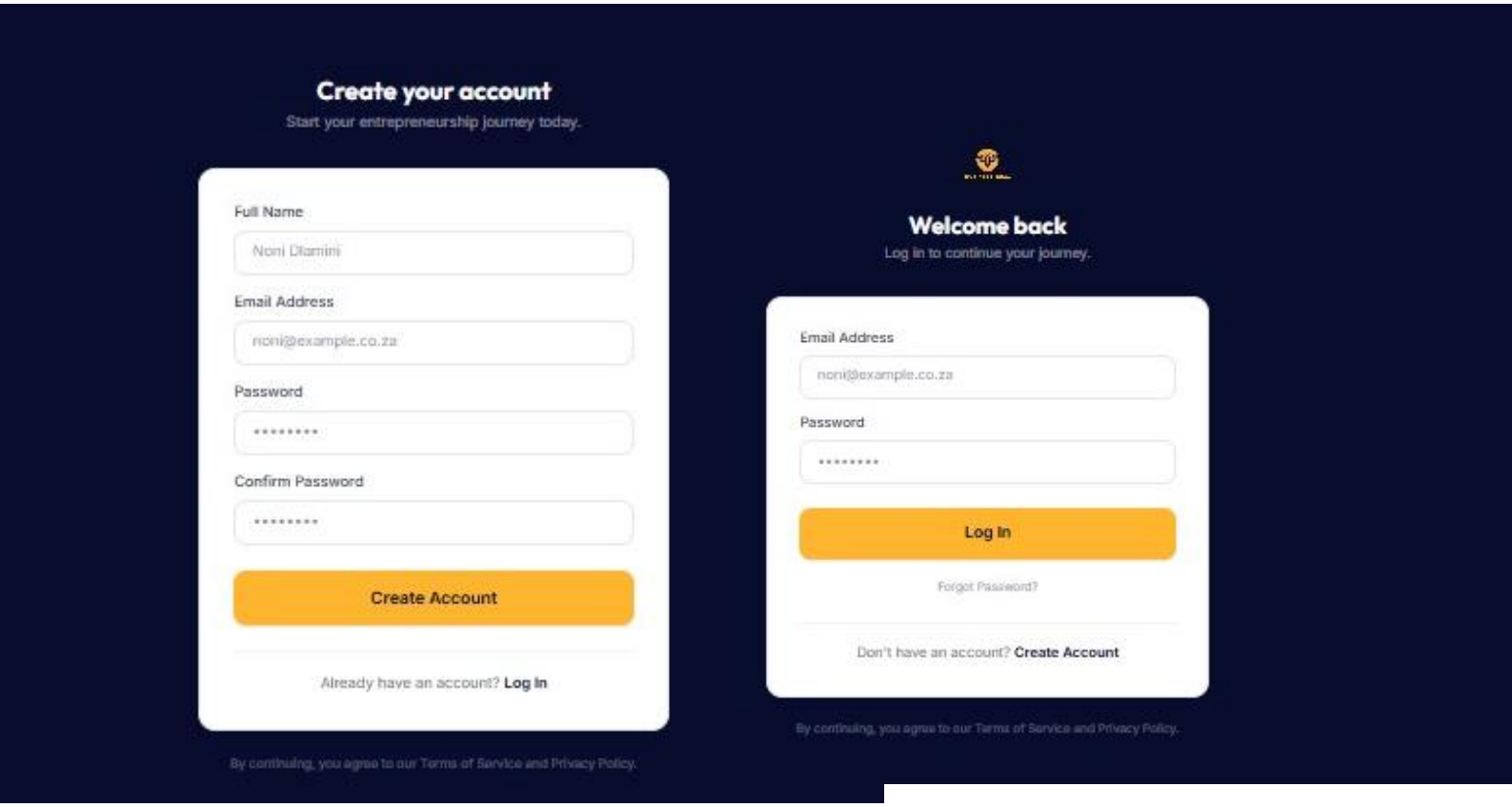


Figure 2.8: Login page and Sign Up Logic

Confirmation Page

Figure 2.9 presents the Business Profile Confirmation Page. This screen allows the entrepreneur to review the business information captured by the system before confirming the profile. The entrepreneur can either edit incorrect information or confirm the details to continue with the business journey.

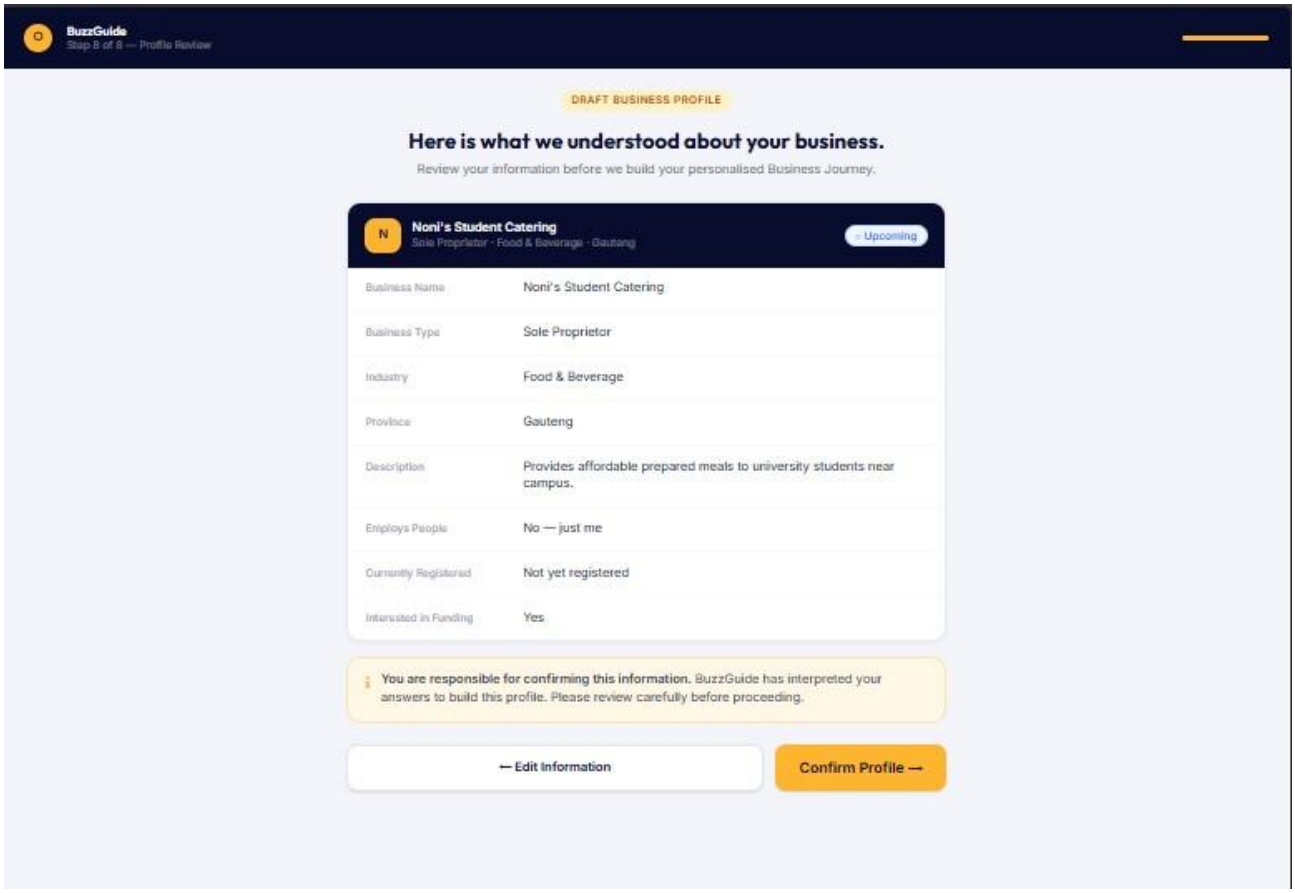


Figure 2.9:The Confirmation Page

Entrepreneur Dashboard

Figure 2.10 presents the Entrepreneur Dashboard. The dashboard provides an overview of the
entrepreneur's business journey, including completed, current and upcoming stages. It also

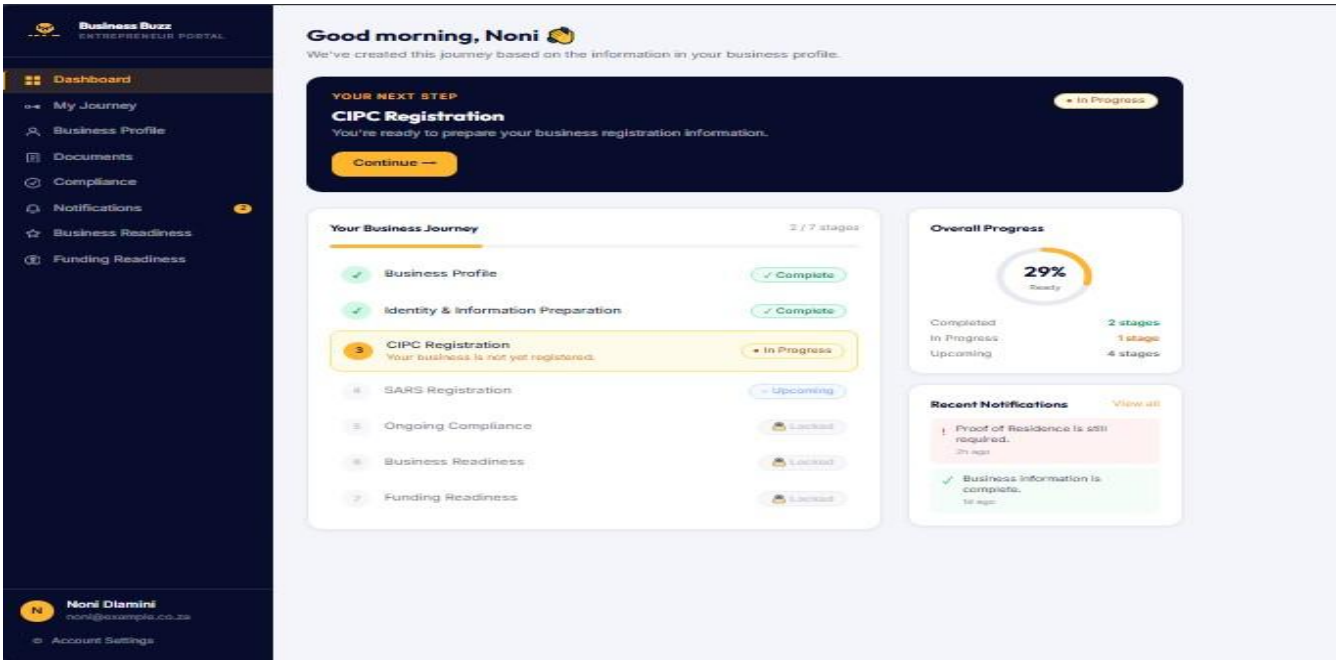


Figure 2.10: :Entrepreneur dashboard

displays. overall progress, recent notifications and the next action that the entrepreneur should complete

1.9.2 Approver Dashboard

Approver Dashboard Page

Figure 2.11 presents the Approver Login Page. This interface allows an authorized Approver to securely access the Approver environment using their registered credentials. Successful authentication provides access to the application review and approval functionality assigned to the Approver role.

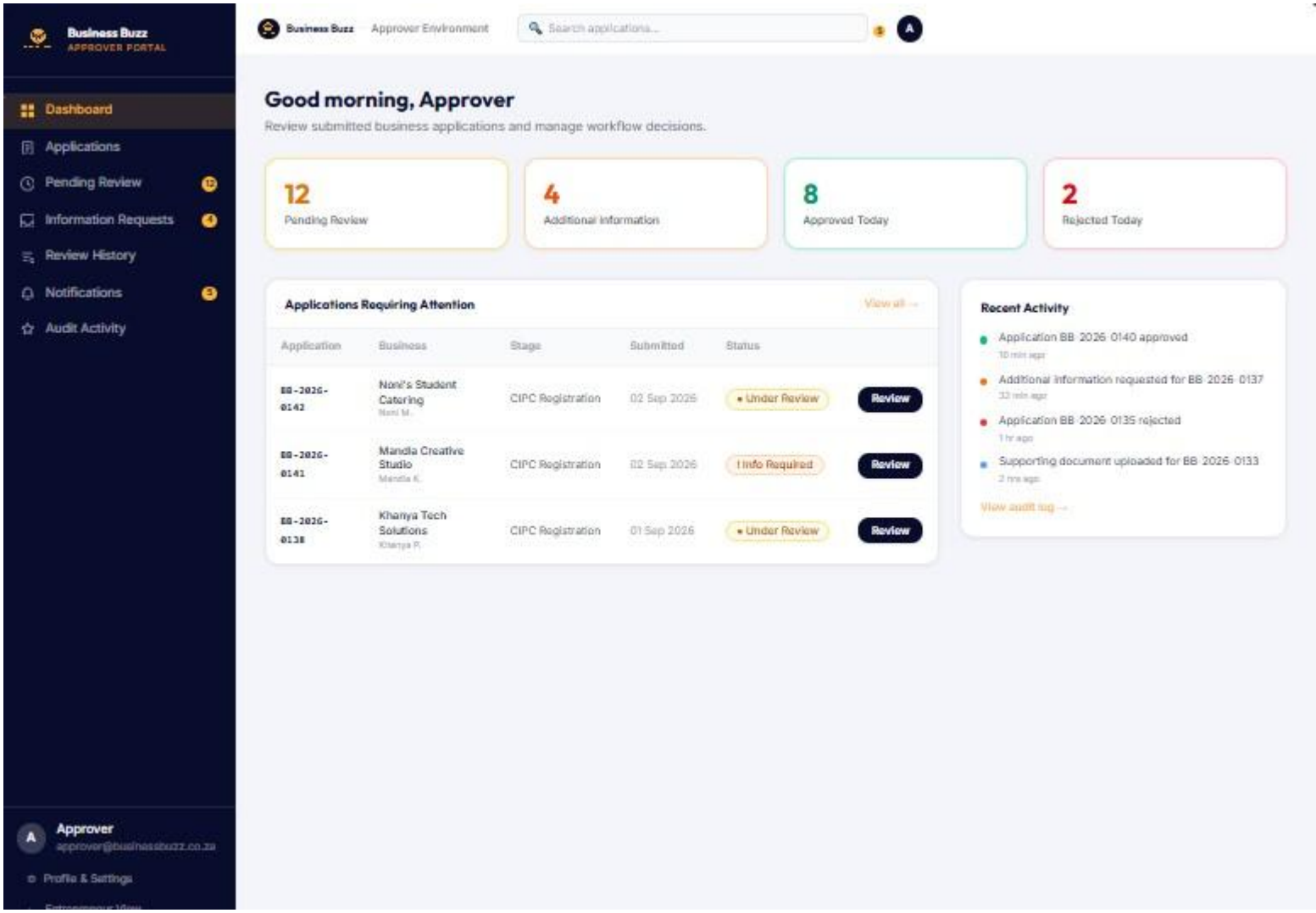


Figure 2.11: Approver dashboard

Bank Dashboard

Figure 2.12 presents the Bank Dashboard. The dashboard provides an overview of businesses available for review, funding enquiries, assessments awaiting attention and completed assessments. It also enables the Bank user to identify businesses requiring assessment and review recent assessment activity.

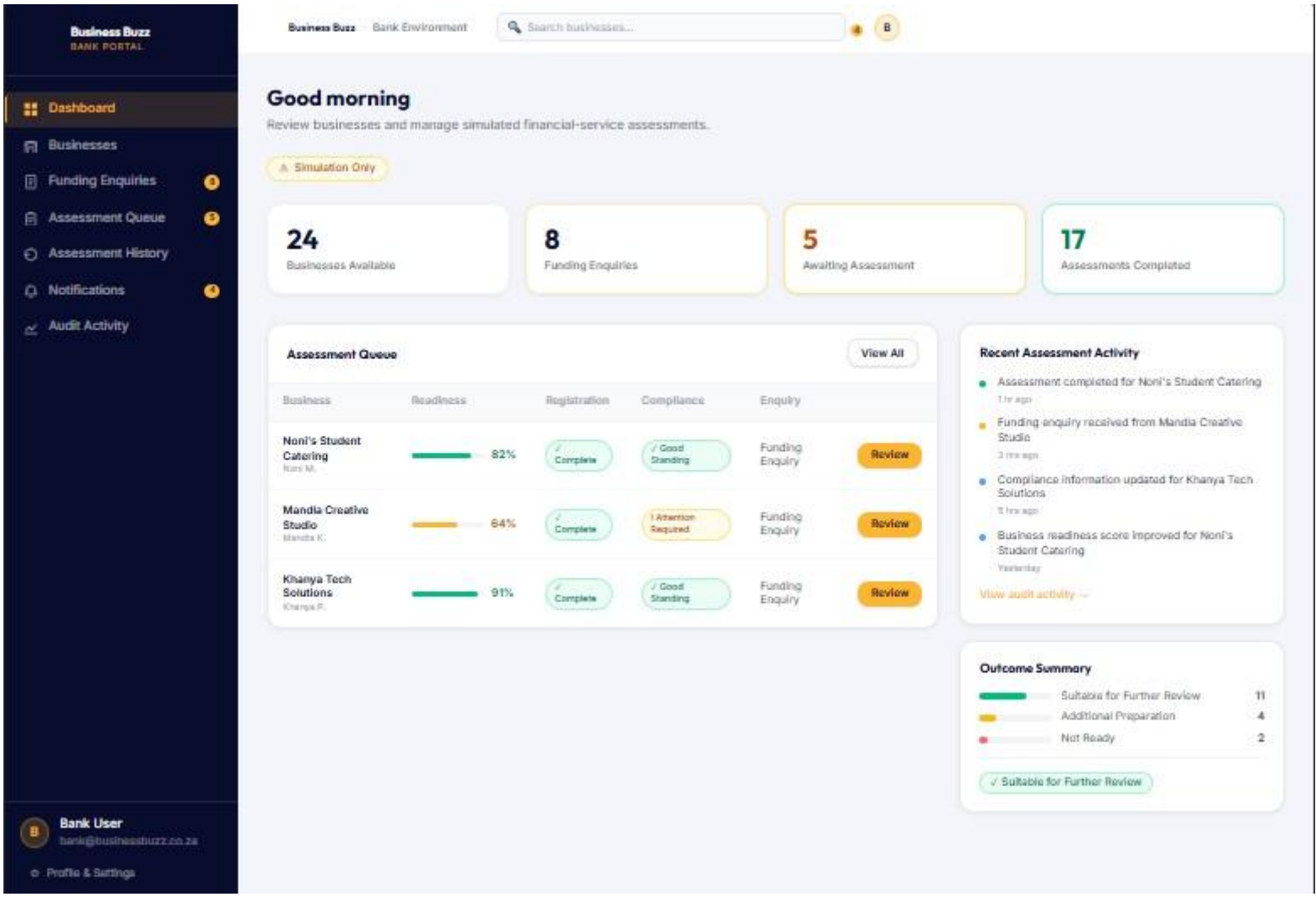


Figure 2.12:Bank Dashboard

3. Implementation

This document describes the implementation of **Business Buzz**, a centralized web-based business registration and compliance management system developed as an academic prototype. The purpose of Business Buzz is to provide entrepreneurs with a single environment in which they can prepare business information, follow simulated registration and compliance processes, generate relevant business documents, and monitor their progress towards business readiness.

The implementation of Business Buzz consists of a web-based frontend, an ASP.NET Core backend, and a Microsoft SQL Server database. The system supports three primary user roles: **Entrepreneur, Approver, and Bank**. Each role is provided with functionality relevant to its responsibilities within the system. The Entrepreneur follows the business registration and compliance journey, the Approver reviews relevant applications and information, and the Bank reviews relevant simulated registration and compliance information when conducting a simulated financial-service assessment.

Business Buzz models business processes associated with entities such as the Companies and Intellectual Property Commission (CIPC), South African Revenue Service (SARS), Unemployment Insurance Fund (UIF), and Home Affairs. These processes are simulated within the academic prototype and do not represent direct integration with official institutional systems.

This document discusses the tools and technologies selected for the implementation of Business Buzz, extracts of significant code developed for the system, external framework and implementation resources used during development, and challenges encountered by the development team.

3.1 Choice of Tools

The Business Buzz application was developed using a combination of development environments, backend technologies, database technologies, frontend technologies, frameworks, and version-control tools. The tools were selected according to the technical requirements of the system and the need for different members of the development team to work on different components of the application in parallel.

3.1.1 Visual Studio 2022

Visual Studio 2022 was used as one of the primary integrated development environments (IDEs) for the Business Buzz project, particularly for frontend development and management of the ASP.NET Core project structure.

Visual Studio 2022 was selected because it provides integrated support for developing .NET applications, including source-code editing, building, debugging, project management, and source-control integration. These capabilities assisted the team in identifying implementation problems and testing the application locally.

According to Microsoft, Visual Studio provides an integrated development environment containing tools for editing, building, debugging, and managing software projects.

3.1.2 Visual Studio Code

Visual Studio Code was also used during the implementation of Business Buzz, particularly for backend development.
Visual Studio Code was selected because it provides a lightweight development environment while supporting C#, .NET, Git, terminal commands, debugging, and extensions.
The use of Visual Studio Code allowed backend development to take place within a preferred development environment while still contributing to the same Business Buzz GitHub repository. This was useful because the frontend and backend were developed by different team members in parallel.
Microsoft describes Visual Studio Code as a source-code editor that supports debugging, source control, extensions, and an integrated development workflow.

3.1.3 ASP.NET Core

ASP.NET Core was selected for the server-side implementation of Business Buzz.
ASP.NET Core provides the framework through which the system handles server-side application functionality, authentication-related operations, session management, application services, and interaction with the application's data layer.

The framework integrates effectively with C# and the .NET platform. Its dependency-injection functionality also allows application services to be registered and made available to the components that require them.

ASP.NET Core was therefore appropriate for implementing the server-side functionality required by Business Buzz while maintaining separation between the user interface, application logic, and data-access components.

3.1.4 C#

C# was selected as the primary programming language for the server-side implementation of Business Buzz.
C# integrates directly with .NET and ASP.NET Core and supports object-oriented programming principles. This enables the application to be organized into structured and reusable entities, services, interfaces, and other components.

Within Business Buzz, C# is used to implement functionality such as authentication, validation, business-profile processing, personalized journey generation, and database interaction.
Microsoft describes C# as a general-purpose programming language within the .NET platform that supports strongly typed and object-oriented application development.

3.1.5 Microsoft SQL Server

Microsoft SQL Server was selected as the relational database management system for Business Buzz.

The database maintains structured information required by the system, including users, business profiles, applications, approval information, funding assessments, journey-related information, and other application records.

A relational database was appropriate because Business Buzz contains multiple related entities and requires relationships and data integrity to be maintained between them.
SQL Server was also suitable because it integrates effectively with the Microsoft and .NET technologies used by Business Buzz and works with Entity Framework Core to support database operations.
Microsoft provides SQL Server as a relational database platform for storing, querying, managing, and securing application data.

3.1.6 Entity Framework Core

Entity Framework Core (EF Core**)** was used as the object-relational mapping framework between the Business Buzz C# application and the SQL Server database.
EF Core enables C# entity classes to represent application data and allows database information to be queried and saved through a DbContext. This reduces the need to manually implement every database interaction directly in SQL.
EF Core migrations were also used during development to update the database structure as the Business Buzz data model evolved. For example, changes to the Business Profile entity could be reflected in the SQL Server database through migrations rather than requiring the entire database to be recreated.
Microsoft (n.d.-e) describes Entity Framework Core as an object-relational mapper for .NET.

3.1.7 HTML5

HTML5 was used to define the structure and content of the Business Buzz frontend.
HTML provides the underlying structure for pages presented to users, including authentication pages, the Entrepreneur Dashboard, profile confirmation, registration information pages, and document-related interfaces.
HTML elements were used to organise headings, forms, navigation controls, buttons, links, sections, and other interface components. This provided the foundation upon which CSS styling and JavaScript behaviour could be applied.
MDN Web Docs describes HTML as the markup language used to structure web pages and their content.

3.1.8 CSS3

CSS3 was used to implement the visual design and presentation of Business Buzz.
Custom CSS controls the application's colour scheme, typography, spacing, sidebar navigation, cards, buttons, status indicators, dashboards, responsive layouts, and other interface elements.
CSS was particularly important for establishing a consistent Business Buzz visual identity across the application. Reusable CSS classes were used across multiple pages to maintain consistency and reduce unnecessary duplication.
According to MDN Web Docs (n.d.-b), CSS is used to describe the presentation and layout of documents written using markup languages such as HTML.

3.1.9 Bootstrap 5.3

Bootstrap 5.3 was used as a frontend toolkit alongside the custom HTML and CSS developed for Business Buzz. The frontend references Bootstrap 5.3.8 through a Content Delivery Network (CDN).

Bootstrap was selected because it provides responsive frontend utilities and reusable styling functionality. It was used together with custom Business Buzz CSS rather than replacing the application's custom visual design.

This allowed the team to benefit from Bootstrap's responsive functionality while maintaining the specific Business Buzz colour scheme, layouts, buttons, navigation elements, and other interface components.

Bootstrap Team (n.d.) describes Bootstrap as a frontend toolkit designed to support responsive and mobile-first web development.

3.1.10 JavaScript

JavaScript was used to provide client-side behaviour and interactivity within the Business Buzz frontend.

JavaScript supports functionality that occurs within the browser, including responding to user-interface events and controlling dynamic frontend behaviour.

During frontend development, JavaScript also supported testing of application interactions and page flows while other server-side functionality was still under development.

MDN Web Docs describes JavaScript as a programming language used to implement functionality and interactive behaviour in web applications.

3.1.11 Git and GitHub

Git and GitHub were used for source-code version control and collaboration between Business Buzz team members.

Git enabled the development team to track changes to the source code and maintain a history of implementation work. GitHub provided the shared remote repository through which team members contributed to the same Business Buzz project.

Separate development branches allowed different components to progress independently without requiring every change to be immediately introduced into the main branch. Changes could be committed and pushed to GitHub and subsequently compared and integrated through the repository workflow.

The Git Project (n.d.) documents Git as a distributed version-control system, while GitHub (n.d.) provides repository-based functionality for storing, managing, and collaborating on source code.

3.2 Extracts of complex code

The following extracts of code are the original work of the Business Buzz development team. They were selected because they represent areas of the system where multiple application requirements had to be combined into working functionality.

3.2.1 Authentication and Password Hashing

The authentication service is responsible for registering users and authenticating existing users. This functionality was considered complex because secure authentication requires more than simply comparing a password with a value stored in the database. It also requires passwords to be handled and stored securely.

During registration, the application first checks whether an account already exists for the supplied email address. A new SystemUser is then created, and the supplied password is converted into a secure password hash using the PasswordHasher<TUser> functionality provided by ASP.NET Core Identity before the value is stored.

The Following extract shows the password hashing process:

```
user.PasswordHash = _passwordHasher.HashPassword(
    user,
    password);
_context.SystemUsers.Add(user);
```

Figure 2.1: Password Hashing

During login, the supplied password is verified against the stored password hash rather than comparing or storing the user's original password.

```
var result = _passwordHasher.VerifyHashedPassword( user,
user.PasswordHash,password);
if (result ==
Microsoft.AspNetCore.Identity.PasswordVerificationResult.Fai
led)
{return null;}
```

Figure 3.2: Password Verification Process

This implementation means that Business Buzz does not need to store users' original passwords. The authentication service also retrieves the user's role during login, allowing the authenticated session to be associated with the appropriate type of system user.
After successful authentication, the application stores the authenticated user's ID and role in the server-side session. This allows subsequent pages to identify the authenticated user rather than relying only on information supplied by the browser. This is consistent with Noni's implemented authentication flow.

3.2.2 Business Profile Creation and Validation

The Business Profile functionality was considered complex because it combines authentication, validation, database verification, and data persistence within one process.

Before a business profile can be created, the application retrieves the authenticated user's ID from the session. This ensures that an entrepreneur cannot create a business profile without first being authenticated.

```
var userId = HttpContext.Session.GetInt32("UserId");
if (userId == null)
{
    return RedirectToPage("/Account/Login");
}
```

Figure 3.3: User Session Management

The application then validates the information entered by the entrepreneur, including the business name, business type, industry, and business location. The selected business type is also verified against the database before the profile is saved.

```
var userId = HttpContext.Session.GetInt32("UserId");
if (userId == null)
{
    return RedirectToPage("/Account/Login");
}
```

Figure 3.3: Server-Side Information Verification

Once the information has passed validation, the profile is associated with the authenticated user.

```
var userId = HttpContext.Session.GetInt32("UserId");
if (userId == null)
{
    return RedirectToPage("/Account/Login");
}
```

Figure 3.4: Linking Onboarding Information to a Business Profile

Associating the profile with the authenticated user allows the Business Profile to become the foundation for later functionality, including the entrepreneur's personalized business journey.

1.2.3 Personalized Journey Generation

The personalized journey functionality was developed so that Business Buzz does not necessarily present every entrepreneur with the same business journey.
The application retrieves the entrepreneur's Business Profile and evaluates information contained within it. The journey service then determines which journey stages are applicable to that particular business.
For example, the EmploysPeople property can determine whether a UIF-related stage should be included, while the SeeksFunding property can determine whether the Funding Readiness stage should be presented to the entrepreneur.

```
var userId = HttpContext.Session.GetInt32("UserId");
if (userId == null)
{
    return RedirectToPage("/Account/Login");
}
```

Figure 3.6: Business Journey Personalization

The journey rules are maintained within the application's infrastructure service rather than being placed directly in the user interface. This separation of responsibilities makes the application easier to maintain and allows the business journey to adapt according to information supplied by the entrepreneur rather than displaying a fixed sequence of stages to every user.

3.3 Source code references (if applicable)

External documentation and framework functionality were consulted during the development of Business Buzz. The application-specific code was developed for the Business Buzz project; however, official framework documentation was used to understand and correctly implement functionality provided by the frameworks and libraries used by the application.

3.3.1 Converting images to byte arrays

Description

Business Buzz uses the Password Hasher<TUser> class provided by ASP.NET Core Identity for password hashing and verification.
The development team did not create a custom password-hashing algorithm. Instead, the Business Buzz authentication service uses framework-provided functionality to generate a hashed representation of a user's password during registration and verify a supplied password during login.
Microsoft documents Password Hasher<TUser> as providing Identity password-hashing functionality, including the HashPassword and VerifyHashedPassword methods.

Usage

The functionality is used within:

- AuthenticationService.RegisterAsync()
- AuthenticationService.LoginAsync()
- User registration functionality
- User login functionality

Microsoft. (n.d.-f). *PasswordHasher<TUser> class*. Microsoft Learn. https://learn.microsoft.com/en-us/dotnet/api/microsoft.aspnetcore.identity.passwordhasher-1

3.3.2 ASP.NET Core Session State

Description

ASP.NET Core session functionality was used to maintain information about an authenticated user between requests.
After successful login, Business Buzz stores the authenticated user's ID and role in the server-side session. Subsequent functionality, such as Business Profile creation, retrieves the user ID from HttpContext.Session to determine which authenticated user is performing the operation.
Microsoft's ASP.NET Core documentation describes session state as a mechanism for maintaining user-specific data while a user browses an application.

Usage

Session functionality is used within:

- Program.cs
- Login functionality
- CreateProfile.cshtml.cs
- Dashboard and user-specific functionality

3.3.3 Entity Framework Core Database Migrations

Description

Entity Framework Core migrations were used to keep the SQL Server database structure synchronized with changes made to the Business Buzz entity model.

For example, when the SeeksFunding property was added to the BusinessProfile entity, a new migration was created and applied to the existing database rather than recreating the database.
EF Core migrations allow a database schema to be incrementally updated as the application's data model changes while preserving existing data.

Usage

Migrations are used within:

- BusinessBuzz.Infrastructure
- EF Core migration files
- Business Profile entity changes
- SQL Server database updates

3.3.4 Entity Framework Core Object-Relational Mapping

Description

Entity Framework Core was used to allow Business Buzz C# entity classes to interact with the relational SQL Server database.
The framework provides object-relational mapping functionality, allowing database data to be queried and saved through C# objects and a DbContext. This functionality provides the basis for database operations throughout the application.
Noni's implementation identifies EF Core as being used for retrieving users, business profiles and journey stages and for saving new business profiles.

Usage

Entity Framework Core is used within:

- BusinessBuzzDbContext
- Authentication Service
- Business Profile functionality
- Journey Service
- Business Buzz entity classes

Microsoft. (n.d.-e). *Entity Framework Core*. Microsoft Learn. https://learn.microsoft.com/en-us/ef/core/

3.3.5 Bootstrap 5.3 Frontend Toolkit

Description

Bootstrap was incorporated into the Business Buzz frontend through its CDN stylesheet to provide responsive frontend utilities and baseline styling functionality.
Bootstrap is used together with custom HTML and CSS developed specifically for Business Buzz. Custom CSS controls the application's project-specific colour scheme, layouts, navigation components, cards, buttons, and other visual elements.

Usage

Bootstrap is referenced within frontend interfaces, including:

- Authentication pages
- Entrepreneur Dashboard
- CIPC Registration page
- Digital Documents page
- Other Entrepreneur portal interfaces

Bootstrap Team. (n.d.). *Get started with Bootstrap*. Bootstrap. https://getbootstrap.com/docs/5.3/getting-started/introduction/

3.4 Problems encountered

This section represents a bit of reflection on your part. Discuss anything in the development of your project which you struggled with. For each problem provide a numbered heading, a description of the problem and how you solved it (eventually). For this section it is fine to write in the first person.

Example:

3.4.1 Creating an Web Application that has an AI Asistant

During the building of the project the one vision as a team we had was the implementation of AI into our project and this implementation apart of the project's main features and it was the first as the backend developer of the group having to develop such a feature and learning different AI concepts that seemed very foreign but with time I finally understood.

I eventually discovered the concept of a token which came in handy when it was time to use Groq for its AI API, though costly it seemed very much worth it for the completion of the vision.

3.4.2 Persistence

As an IT student , the concept of persistence is something that is drilled into you especially if you are coding and making sure a feature you implemented works on more than one occasion and I took it for granted given that the projects we had were very easy and therefore persistence wasn't something I would check for cause the code ran but having to build an application that could have real world implementation I had to make sure that every part I code because I had to account for something going wrong and working well. This came in the form of looking at the database and take note oof any visual changes to last through testing.

3.4.3 Developing a methodology to my coding

As a student in IT you encounter and get taught about different concepts with the expectation that you will understand and approaching coding this year I had to develop my "own" methodology adapted from the Agile Methodology and with the different deadlines and ever-changing requirements I was fortunate that I had knowledge that helped me very much.
Having a personal timetable and objectives I helped with the development process abit easier.

4. Conclusion

The development of BusinessBuzz by BuzzTech Solutions aimed to address the challenges faced by aspiring South African entrepreneurs when starting and managing their businesses. These challenges include complex business registration procedures, regulatory compliance requirements, document management and limited access to funding information. The project sought to provide a centralised, user-friendly digital platform that simplifies these processes and guides entrepreneurs from the initial business idea towards business readiness.

The project achieved significant progress through the development of a functional web-based prototype consisting of three main portals: the Entrepreneur Portal, Application Approver Portal and Bank Portal. The Entrepreneur Portal supports business profile management, CIPC and SARS registration preparation, compliance tracking, document management, business readiness assessments and funding enquiries. The Approver Portal facilitates application reviews, document verification and decision-making, while the Bank Portal supports funding enquiry reviews and simulated funding decisions. Together, these features demonstrate how digital technology can improve the organisation, accessibility and transparency of business preparation processes.

Throughout the development process, the team gained valuable experience in software development, system design, problem-solving, collaboration, version control and functional testing. The project highlighted the importance of developing user-friendly interfaces, maintaining data consistency and ensuring that different system components work together effectively. It also reinforced the need for secure data management and reliable system integration.

Although BusinessBuzz has demonstrated its core functionality, further improvements are necessary to enhance its reliability and scalability. Future development will focus on integrating the ASP.NET Core backend with Microsoft SQL Server, implementing secure authentication and role-based access control, improving document security and introducing automated notifications. Potential future integration with authorised government and financial institutions could further expand the platform's capabilities.

In conclusion, BusinessBuzz demonstrates the potential of technology to simplify business preparation and support entrepreneurship in South Africa. While the current system remains an academic prototype, it provides a foundation for a more comprehensive digital business support platform. The project has successfully demonstrated its proposed concept while equipping the development team with practical technical skills and valuable experience in addressing real-world challenges through innovative software solutions.

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References

Companies and Intellectual Property Commission. (n.d.-a). CIPC eServices. https://eservices.cipc.co.za/
Companies and Intellectual Property Commission. (n.d.-b). BizPortal. https://www.bizportal.gov.za/
Companies and Intellectual Property Commission. (n.d.-c). FAQs. https://www.cipc.co.za/?page_id=4160
Myburg, J., & Schutte, D. (2025). SMMEs' business mindset on tax compliance in North West: Perceptions and challenges. Southern African Journal of Entrepreneurship and Small Business Management, 17(1), a1085. https://doi.org/10.4102/sajesbm.v17i1.1085
Nieuwenhuizen, C. (2019). The effect of regulations and legislation on small, micro and medium enterprises in South Africa. Development Southern Africa, 36(5), 666–677. https://doi.org/10.1080/0376835X.2019.1581053
Organisation for Economic Co-operation and Development. (2025). OECD economic surveys: South Africa 2025. OECD Publishing. https://doi.org/10.1787/7e6a132a-en
Organisation for Economic Co-operation and Development. (2026). Foundations for growth and competitiveness 2026. OECD Publishing. https://doi.org/10.1787/40a7532f-en
Sibiya, A., van der Westhuizen, J., & Sibiya, B. (2023). Challenges experienced by SMMEs and interventions by the South African national and provincial government: A literature review. African Journal of Inter/Multidisciplinary Studies, 5(1), 1–11. https://doi.org/10.51415/ajims.v5i1.1224
South African Revenue Service. (2025). Guide to submit your individual income tax return via eFiling: External guide. https://www.sars.gov.za/wp-content/uploads/Ops/Guides/IT-AE-36-G06-Guide-to-submit-your-individual-income-tax-return-via-eFiling-External-Guide.pdf
Unemployment Insurance Fund. (n.d.). Basic guide to the Unemployment Insurance Fund (UIF). Department of Employment and Labour. https://www.labour.gov.za/DocumentCenter/Publications/Unemployment%20Insurance%20Fund/Basic%20Guide%20to%20the%20Unemployment%20Insurance%20Fund%20%28UIF%29.pdf

Bootstrap Team. (n.d.). *Get started with Bootstrap*. Bootstrap. https://getbootstrap.com/docs/5.3/getting-started/introduction/
Git Project. (n.d.). *Git documentation*. https://git-scm.com/docs/git
GitHub. (n.d.). *Repositories documentation*. GitHub Docs. https://docs.github.com/en/repositories
MDN Web Docs. *HTML: HyperText Markup Language*. Mozilla. https://developer.mozilla.org/en-US/docs/Web/HTML
MDN Web Docs. *CSS: Cascading Style Sheets*. Mozilla. https://developer.mozilla.org/en-US/docs/Web/CSS
MDN Web Docs. (n.d.-c). *JavaScript guide*. Mozilla.
Microsoft. (n.d.-a). *What is Visual Studio?* Microsoft Learn. https://learn.microsoft.com/en-us/visualstudio/get-started/visual-studio-ide

Microsoft. (n.d.-c). *C# documentation*. Microsoft Learn. https://learn.microsoft.com/en-us/dotnet/csharp/
Microsoft. (n.d.-d). *SQL Server technical documentation*. Microsoft Learn. https://learn.microsoft.com/en-us/sql/sql-server/
Microsoft. (n.d.-e). *Entity Framework Core*. Microsoft Learn. https://learn.microsoft.com/en-us/ef/core/
Microsoft. (n.d.-f). *PasswordHasher<TUser> class*. Microsoft Learn. https://learn.microsoft.com/en-us/dotnet/api/microsoft.aspnetcore.identity.passwordhasher-1
Microsoft. (n.d.-g). *Session and state management in ASP.NET Core*. Microsoft Learn. https://learn.microsoft.com/en-us/aspnet/core/fundamentals/app-state
Microsoft. (n.d.-h). *Migrations overview*. Microsoft Learn. https://learn.microsoft.com/en-us/ef/core/managing-schemas/migrations/

A. Business case revisions

This document reports on the changes made to the business case, after assessment.

A.1 Changes made in response to comments

A.1.1 Justify (align) document text

Commentreceived:
The lecturer requested that the document text be justified (aligned) rather than left-aligned.
Changes made
All body text in the Business Case has been fully justified (aligned to both left and right margins). This applies to all paragraphs, bullet points, and table content throughout the document. Headings remain left-aligned for consistency and readability.
Sections affected:

- Section 1.1 (The Problem)
- Section 1.2 (Project Objectives)
- Section 1.3 (Problem Background / Literature Review)
- Section 1.4 (Related Systems Analysis)
- Section 1.5 (The Project Plan)
- Section 1.6 (Risk Analysis)

A.2 Changes made as the project progressed

A.2.1 Changed objectives

Commentreceived:
The lecturer requested that the Project Objectives section should indicate **how** each objective will be achieve
Changes made

Section 1.2 (Project Objectives) was restructured. Each objective now includes:

- A clear statement of **what** the objective is
- A clear statement of **how** it will be achieved

This was presented in terms of a descriptive paragraph on each objective

B. System requirements, specifications and technical design revisions

B.1 Changes made in response to comments

B.2 Changes made as the project progressed

B.2.1 Extra feature added

Added Focal Point to Limit Scope
Original approach:
The original System Requirements document targeted SMMEs broadly without a specific focal point.

Change made:
A focal point was added to the document, defining the target segment as micro and very small enterprises in the pre-registration phase.
Reason for change:

- The original scope was too broad for the academic timeframe
- A well-defined target segment makes the system's conditional workflow logic more relevant
- It aligns with the problem statement and Business Case revisions

Sections affected:

- Section 1.2 (Project Scope)
- Section 1.3 (Business Requirements)

C. Implementation revisions

This document reports on the changes made to the business case, after assessment.

C.1 Changes made in response to comments

C.1.1 Extract of complex code was boilerplate

```
document.getElementById("generateReadiness")
    .addEventListener("click", function () {

        // Read the entrepreneur's saved business profile.
        let profile = {};
        let fundingEnquiry = null;

        try {
            profile = JSON.parse(
                localStorage.getItem("businessBuzzBusinessProfile")
            ) || {};
        } catch (error) {
            console.error("Unable to read business profile:", error);
        }

        // Funding enquiry is saved in sessionStorage
        // by your existing funding-readiness.js.
        try {
            fundingEnquiry = JSON.parse(
                sessionStorage.getItem("businessBuzzFundingEnquiry")
            );
        } catch (error) {
            console.error("Unable to read funding enquiry:", error);
        }

        const display = value =>
            value === undefined ||
                value === null ||
                String(value).trim() === ""
                ? "Not provided"
                : String(value);

        // These requirements match the existing
        // business-readiness.js demonstration checklist.
        const businessRequirements = [
            { name: "Business Profile", complete: true },
            { name: "CIPC Registration Preparation", complete: true },
            { name: "Business Documentation", complete: true },
            { name: "SARS Compliance Preparation", complete: false },
            { name: "Compliance Review", complete: false }
        ];

        // These requirements match the existing
        // funding-readiness.js demonstration checklist.
        const fundingRequirements = [
            { name: "Business Profile", complete: true },
            { name: "Registration Preparation", complete: true },
            { name: "Tax Compliance Preparation", complete: false },
            { name: "Financial Information", complete: false },
            { name: "Business Readiness Review", complete: false }
        ];
```